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Review of medical image processing using quantum-enabled algorithms

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Abstract
Efficient and reliable storage, analysis, and transmission of medical images are imperative for accurate diagnosis, treatment, and management of various diseases. Since quantum computing can revolutionize big data analytics by providing faster solutions and security tactics, numerous studies in this field have focused on the use of quantum and quantum-inspired algorithms to enhance the performance of traditional medical image processing approaches. This review aims to provide readers with a succinct yet adequate compendium of the advances in medical image processing combined with quantum behaviors for disease diagnosis and medical image security. Some open challenges are outlined, identifying the performance limitations of current quantum technology in their applications, while addressing the short-, medium-, and long-term development plans of this field in designing future quantum healthcare systems. We hope that this review will provide full guidance for upcoming researchers interested in this area and will stimulate further appetite of experts already active in this area aimed at the pursuit of more advanced quantum paradigms in medical image processing applications.

Keywords Biomedical engineering · Machine learning · Quantum computing · Medical image processing · Computer-aided diagnosis · Medical image security

1 Introduction

With the ongoing advancement of medical devices and technology, various imaging approaches can provide medical images of human body organs with detailed anatomical information. This has led to medical images that are widely used to diagnose different diseases, such as Alzheimer's disease (Chen et al. 2023), pulmonary nodules (Mei et al. 2022), breast masses (Yan et al. 2021), and cervical cancer (Castle et al. 2021). Therefore, medical images are one of the primary methods and grounds for clinical diagnosis in the contemporary medical field. However, there may be some problems with the process, as medical images are used to diagnose diseases. For example, radiologists must diagnose a large quantity of images daily, which can yield missed diagnoses owing to visual fatigue or a lack of experience (Schmauch et al. 2019; Cha et al. 2019; Wang et al. 2020). More importantly, massive numbers of medical images are transmitted over

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public networks because of the demand for telemedicine, which potentially facilitates the manipulation, destruction, or theft of medical images without authorization (Yan et al. 2022). These undesirable phenomena result in poor treatment effects and severe economic losses. To solve these problems, medical image processing and artificial intelligence technologies have been developed and used to help physicians diagnose diseases and to improve their efficiency, accuracy, and reliability (Anter et al. 2021; Ström et al. 2020; Kumar et al. 2023). In addition, research on security technologies (e.g., cryptography and information hiding) in medical image processing can effectively provide safeguards for the transmission of medical images by verifying the integrity and authenticity of the image (Magdy et al. 2022). However, owing to the ongoing growth of a large amount of medical data that needs to be processed and analyzed daily, it is becoming a challenge to effectively accomplish these tasks in the case of limited computational devices and configurations. Providing nontraditional algorithms for highly efficient and secure medical image processing is an urgent concern that needs to be addressed.

1.1 Main motivations of utilizing quantum methods in medical image processing

Conventional algorithms frequently exhibit high computational complexity and time consumption, posing significant challenges when dealing with high-dimensional data and other complex scenarios. Quantum information science is an interdisciplinary field that involves physics, mathematics, and computer science and is devoted to finding methods to exploit the principles of quantum mechanics in nature, such as superposition and entanglement, to perform information processing beyond the limit of conventional computation (Yan and Venegas-Andraca 2020; Biamonte et al. 2017). Therefore, quantum methods are often used to outperform, enhance, or complement classical algorithms, particularly to solve specific problems related to computing speed and tamper-proof security (Wiebe et al. 2012; Montanaro 2016; Yan et al. 2023). Notably, a growing number of researchers are focusing their efforts on harnessing quantum methods to tackle the intricate challenges in medical image processing. Their aim is to enhance the efficiency and security of tasks in this field, especially given the limitations of constrained computational resources and configurations. The intrinsic driving force primarily stems from the potential advantages that quantum computing could bring to the medical image processing domain, such as:

(1) Increased computational power: Quantum computing has great potential to perform certain computations in parallel, which allows them to possibly tackle multivariate data in medical image processing much more efficiently than classical approaches.
(2) Enhanced accuracy: Quantum algorithms can leverage the quantum mechanical effect to avoid getting stuck in local minima when optimizing complex functions, which might be crucial for tasks like medical image segmentation and classification.
(3) Improved security: Quantum states adhere to the principles of uncertainty and no-cloning, which could likely enable the use of quantum methods in medical image processing to improve the security of medical data for telemedicine applications.
(4) Handling quantum effects: In certain medical imaging modalities, such as magnetic resonance imaging (MRI) or quantum imaging techniques, the underlying physical processes involve quantum mechanical phenomena. Quantum approaches might be

better equipped to model and account for these quantum effects, potentially leading to enhanced image quality and diagnostic capabilities.

To achieve these goals, researchers interested in this cross-disciplinary field have been exploring various approaches. Technically, these pioneering efforts and the subsequent related studies can be classified into three main groups:

(1) Quantum medical image processing: These applications derive inspiration from the expectation that quantum devices will be achieved in a short time, resulting in such research focusing on expanding medical image processing tasks and applications to the quantum computing framework. For example, a quantum representation was formulated for the medical image and a secure transmission scheme was designed based on it (Janani and Brindha 2021).
(2) Quantum medical imaging: These applications utilize quantum-correlated imaging (ghost imaging) to perform real-time online imaging for disease diagnosis and treatment (Zhang and Yang 2017). It focuses on devising novel techniques for medical imaging at the quantum level by utilizing the spatial properties of the quantum states of light or spatial quantum correlations.
(3) Quantum-assisted medical image processing: These applications seek to leverage quantum and quantum-inspired algorithms to enhance the implementation of various established tasks and applications within the realm of medical image processing. For example, a quantum fruit fly algorithm was used to select the optimal features of medical image data to improve diagnosis accuracy (Nijaguna et al. 2023).

However, the research in groups (1) and (2) is heavily reliant on quantum computing devices, which are currently limited to theoretical explorations due to the incomplete development and popularization of universal and large-scale quantum computers. Therefore, groups (1) and (2) are considered outside the scope of this review, and we concentrate on group (3) to provide an overview of the latest advances in medical image processing based on quantum algorithms and quantum-inspired algorithms. It is believed that this will not only be beneficial for the development of quantum-assisted medical image processing but will also provide an impetus and reference value for the advancement of the other two fields.

1.2 Comparisons with related review studies

As illustrated in Fig. 1a, interdisciplinary research encompassing the domains of quantum algorithms, medical care, and image processing has produced various notable research achievements. Basically, research involving these fields can be divided into four directions using the famous three circles model: medical care and image processing (M-I), image processing and quantum algorithms (I-Q), medical care and quantum algorithms (M-Q), as well as medical care, image processing, and quantum algorithms (M-I-Q). As presented in Table 1, this study is compared with relevant review papers, including the four subareas aforementioned. Most of the reviews presented in Table 1 are in the M-I category and focus on discussing image processing based on traditional (nonquantum) algorithms, which are mainly involved in the fields of disease diagnosis (Litjens et al. 2017; Ker et al. 2017; Anwar et al. 2018; Maier et al. 2019; Cheplygina et al. 2019; Singh et al. 2020) and information security (Magdy et al. 2022; Gull and Parah 2024). In addition, some review papers

belong to the I-Q category and converge on the representation and analysis of images using quantum algorithms for security techniques aiming to improve image security (Yan et al. 2017; Min-Allah et al. 2022; Nouioua and Belbachir 2023). However, reviews of the M-Q category that discuss the application of quantum algorithms in the healthcare field are few, including drug discovery, personalized medicine, and information security (Rasool et al. 2023). Only the M-I-Q category includes quantum algorithms that are applied to medical image-based healthcare.

We conducted a succinct analysis of the M-I-Q category and the corresponding percentage of published papers according to the Web of Science database from 2014 to date, as shown in Fig. 2a. Undoubtedly, the development of quantum-enabled algorithms for medical image processing has exhibited steady growth over the past few years. Furthermore, as shown in Fig. 2b, we note that M-I-Q research mainly involves two subtopics: disease diagnosis (48.89%) and medical image security (26.06%). In the M-I-Q category, Wei et al. reviewed a paper on quantum machine learning-based medical image analysis, which was a valuable first review of applications using quantum machine learning algorithms in the field of medical image (Wei et al. 2023). However, the review focused more on summarizing medical image segmentation and classification without systematically discussing quantum and quantum-inspired algorithms and their applications in terms of medical image preprocessing and security technologies.

In summary, unlike previous review papers, we comprehensively introduce for the first time an overview of the progress of medical image processing-based disease diagnosis and security techniques using quantum and quantum-inspired algorithms over the past decades. The diagram presented in Fig. 1b highlights the central and principal areas that are the main focus of our review.

1.3 Primary concerns in the field and main contributions of this review

As this emerging field rapidly advances, researchers immersed in quantum-assisted medical image processing may reasonably have several relevant concerns worth considering. Key questions that could guide further discussion and exploration in this review include:

(1) What are the quantum-enabled algorithms that are being investigated, and how can medical image processing tasks stand to benefit from them?

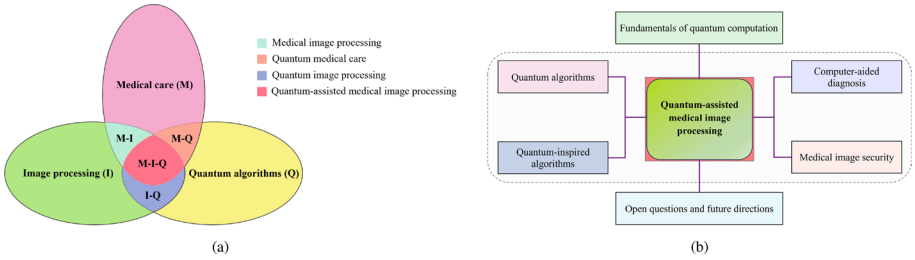


Fig. 1 **a** Illustration of related studies using three circles model. **b** Outline of the review on quantum-assisted medical image processing

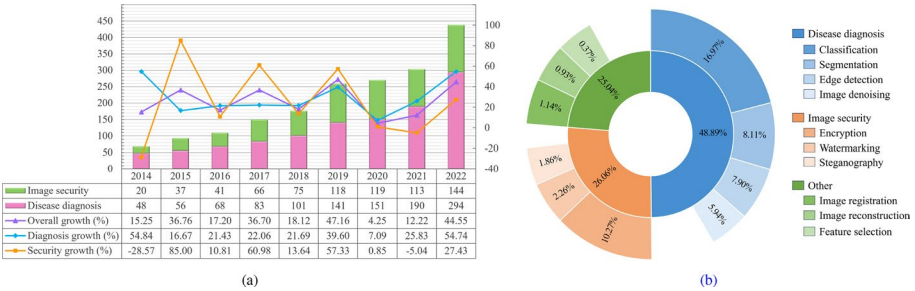


Fig. 2 **a** Bar graph and corresponding table data showing the number of papers published since 2014 and percentile change each year relative to the previous year. **b** Sunburst chart that provides a rough summary and classification of available results in the field

(2) What are the specific applications of quantum-enabled algorithms in medical image processing, and what are the associated advantages they can provide?
(3) What are the existing limitations, key open questions, and potential future directions of quantum-assisted medical image processing domain?

The primary objective of this review is to offer readers a comprehensive yet concise overview of the current state of the quantum-assisted medical image processing field. To this end, the key contributions of this review are summarized as follows:

(1) The foundational principles of quantum computing and the technical specifications of several widely used quantum-enabled algorithms are outlined, along with a broad discussion on leveraging these approaches for medical image processing applications.
(2) Current findings in preliminary processing, segmentation, and classification, and critical steps involved in medical image-based disease diagnosis using quantum-enabled algorithms are systematically analyzed.
(3) In the context of medical image security, techniques to protect patient privacy, such as encryption, watermarking, and steganography, using quantum-enabled algorithms have been extensively explored.
(4) Emerging trends, open challenges, and future research directions in quantum-assisted medical image processing are discussed to address existing limitations and facilitate further advancements in the field.

The rest of this review is expanded along with the main contributions aforementioned as follows. An introduction to quantum computing, quantum algorithms, and quantum-inspired algorithms is presented in Sect. 2. Medical image-based disease diagnosis and security techniques using quantum-enabled algorithms are discussed in Sects. 3 and 4. Finally, discussions and conclusions regarding the development of quantum-assisted medical image processing are outlined in Sect. 5.

Table 1 A comparative analysis of this review and related studies

Category	References	Year	Medical care	Image processing	Quantum algorithms	Quantum-inspired algorithms	Disease diagnosis	Information security
M-I	Litjens et al. (2017)	2017	✓	✓	×	×	✓	×
	Ker et al. (2017)	2017	✓	✓	×	×	✓	×
	Anwar et al. (2018)	2018	✓	✓	×	×	✓	×
	Maier et al. (2019)	2019	✓	✓	×	×	✓	×
	Cheplygina et al. (2019)	2019	✓	✓	×	×	✓	×
	Singh et al. (2020)	2020	✓	✓	×	×	✓	×
	Magdy et al. (2022)	2022	✓	✓	×	×	×	✓
	Gull and Parah (2024)	2023	✓	✓	×	×	×	✓
I-Q	Yan et al. (2017)	2017	×	✓	✓	×	×	✓
	Min-Allah et al. (2022)	2022	×	✓	✓	×	×	✓
	Nouioua and Belbachir (2023)	2023	×	✓	✓	×	×	✓
M-Q	Rasool et al. (2023)	2023	✓	×	✓	×	×	✓
M-I-Q	Wei et al. (2023)	2023	✓	✓	✓	×	✓	×
	This review	2024	✓	✓	✓	✓	✓	✓

Note that the symbol ✓ represents the point included in the review, whereas the symbol × indicates the point not included or comprehensively overviewed in the review

2 Quantum computing and quantum-enabled algorithms

Quantum computing leverages the laws of quantum mechanics to address problems that are too large or intricate for conventional computers to solve. The prerequisite for understanding the advantages of quantum computing is to grasp its fundamental concepts and principles. Therefore, in this section, the fundamentals of quantum computing is outlined in order to help readers better understand quantum and quantum-inspired algorithms, especially their application to medical image processing.

2.1 Background theory of quantum computing

Concepts of quantum bits, registers, gates, and circuits are sequentially introduced wherein some important principles such as superposition, entanglement, parallelism, and no-cloning theorem are illustrated. Moreover, quantum-enabled algorithms, including both quantum and quantum-inspired algorithms, are defined and differentiated as they are applied in this review.

(1) *Quantum bits*

In conventional computers, information units are typically represented using a "0" or "1" bit in binary format. However, in the quantum field, computing units surpass this constraint, demonstrating a remarkable capability to exist not solely in the states of "0" or "1" but also in a superposed state (called a qubit). This is a fundamental principle in quantum mechanics known as quantum superposition. In addition, the representation of quantum states is commonly performed using the *bra-ket* notation, also known as the Dirac notation (Yan et al. 2014). This notation utilizes vertical bars and angle brackets to denote quantum states. Within this symbolic system, each quantum state is portrayed as a vector in Hilbert space, referred to as a *ket*:

$$|\mu\rangle = \begin{pmatrix} \mu_0 \\ \mu_1 \\ \vdots \\ \mu_{n-1} \end{pmatrix}, \ \mu_i \in C, \ i = 0, 1, \ldots, n-1. \tag{1}$$

The adjoint of the *ket* is referred to as *bra*, which is defined as follows:

$$\langle\mu| = |\mu\rangle^{\dagger} = (\mu_0^* \ \ \mu_1^* \ \ \ldots \ \ \mu_{n-1}^*), \tag{2}$$

where u^* is the notation for the complex conjugate of u.

In analogy to the definition of a bit in classical computation, a qubit, which is the fundamental unit of information in a quantum system, is considered a unit vector in a two-dimensional Hilbert space. It is defined by

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \alpha|0\rangle + \beta|1\rangle, \tag{3}$$

where α and β denote probability amplitudes, which are complex numbers. Note that α and β must satisfy $|\ \alpha\ |^2 + |\ \beta\ |^2 = 1$. The states $|0\rangle$ and $|1\rangle$ are referred to as computational basis states that form an orthonormal basis for this vector space. If one lets $x \in \{0, 1\}$, then $\langle x|$ is the conjugate transpose of $|x\rangle$ that has two components in two-dimensional Hilbert space: $\langle 0| = (1 \ \ 0)$ and $\langle 1| = (0 \ \ 1)$. If one lets $y \in \{0, 1\}$, then $\langle x|y\rangle$, i.e., $\langle x||y\rangle$, is the inner

product of $|x\rangle$ and $|y\rangle$, where the result is a scalar. In addition, $|x\rangle\langle y|$ denotes the outer product of $|x\rangle$ and $|y\rangle$, which is an operator in matrix form. If $|0\rangle\langle 0|$ is operated on $|\psi\rangle$, the result of $\alpha|0\rangle$ is acquired, which indicates that $|0\rangle\langle 0|$ has extracted the $|0\rangle$ component from $|\psi\rangle$, or that $|0\rangle\langle 0|$ projects $|\psi\rangle$ onto $|0\rangle$, and $|\psi\rangle$ is measured in the $|0\rangle$ direction. Similarly, the result of $\beta|1\rangle$ can be obtained, which indicates that $|1\rangle\langle 1|$ projects $|\psi\rangle$ onto $|1\rangle$, and $|\psi\rangle$ is measured in the $|1\rangle$ direction. Quantum measurements are a specialized procedure used to convert the transformed quantum information into classical information. Although a qubit can represent a superposition of multiple quantum states, upon measurement the observed outcome is only either 0 or 1, with a specific probability distribution.

(2) *Quantum registers*

A quantum register, analogous to the classical processor register, is a system comprising multiple qubits (Dutt et al. 2007). Quantum computers perform computations by exploiting qubits contained within a quantum register. To indicate the composition of quantum systems, the tensor product (denoted as $\otimes$) is employed. Assume that A and B matrices denote:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} \text{ and } B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1q} \\ b_{21} & b_{22} & \cdots & b_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \cdots & b_{pq} \end{pmatrix}. \tag{4}$$

The tensor product of the two matrices can be represented as follows:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1m}B \\ a_{21}B & a_{22}B & \cdots & a_{2m}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nm}B \end{pmatrix}. \tag{5}$$

In addition, the tensor product of matrix A for n times is expressed as $A^{\otimes n} = A \otimes A \otimes \cdots \otimes A$. Similarly, for two vectors, $|u\rangle$ and $|v\rangle$, their tensor product can be denoted as $|u\rangle \otimes |v\rangle$, which is shortened to $|u\rangle|v\rangle$ or $|uv\rangle$. Assume that there are two qubits in a quantum register, the two-qubit system contains four computational basis states, represented as $|00\rangle, |01\rangle, |10\rangle$, and $|11\rangle$. A pair of qubits may also exist in superpositions of these four states, thereby entailing that the quantum state of two qubits concerns the assignment of a complex coefficient to each computational basis state, such that

$$|\varphi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle. \tag{6}$$

Analogous to the scenario involving a single qubit, the measurement outcomes 00, 01, 10, and 11 are observed with corresponding probabilities $|\alpha_{00}|^2, |\alpha_{01}|^2, |\alpha_{10}|^2, |\alpha_{11}|^2$, and satisfies $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$. For a two-qubit system, a subset of the qubits can be measured, such as the first qubit. Measuring the first qubit alone gives 0 with probability $|\alpha_{00}|^2 + |\alpha_{01}|^2$, resulting in the post-measurement state:

$$|\varphi'\rangle = \frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} \tag{7}$$

Notably, when the state of multiple qubits cannot be represented by a tensor product, it is referred to as an entangled state. For example, $|\varphi\rangle_A = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $|\varphi\rangle_B = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ are in entangled states. This is an essential principle in quantum mechanics, that is, quantum entanglement. Entanglement signifies a unique and

inseparable correlation between the quantum bits, where the measurement or manipulation of one qubit instantaneously affects the state of the other qubits regardless of their physical separation. For instance, upon measuring the state of $|\varphi\rangle_A$, the state of the other qubit will collapse to the same state, whereas measuring the state of $|\varphi\rangle_B$ will result in the other qubit assuming the opposite state.

(3) *Quantum gates*

Quantum gates, alternatively referred to as quantum logic gates, are employed in quantum computing to implement logical operations on quantum registers (Postler et al. 2022). Assume that an operator U_f on a quantum state is a unitary matrix, i.e.,

$$U_f U_f^{\dagger} = I, \tag{8}$$

where I is an identity matrix, when imposed on a single qubit can be described as

$$I = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{9}$$

In addition, $U_f^{\dagger}$ is the matrix that results from taking the conjugate transpose of U_f, both of which are indicated as

$$U_f = \begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix} \text{ and } U_f^{\dagger} = \begin{pmatrix} a_{00}^* & a_{01}^* \\ a_{10}^* & a_{11}^* \end{pmatrix}, \tag{10}$$

where a_{ij}^* is a complex conjugate of a_{ij}, $i,j \in \{0,1\}$. The quantum transform, also known as a unitary transform, employs a unitary matrix as the operator for quantum operations. The unitary transform can ensure that the transformed qubit remains in its normalized state. Due to the reversibility of the unitary transform, the input and output states can be interconverted by using U_f and $U_f^{\dagger}$ transforms, i.e.,

$$U_f|x\rangle = |f(x)\rangle, \quad U_f^{\dagger}|f(x)\rangle = |x\rangle. \tag{11}$$

Note that quantum parallelism enables quantum computers to evaluate a function $f(x)$ simultaneously for multiple values of x. The common quantum gates include the quantum NOT gate, Hadamard gate, controlled-NOT gate, and Toffoli gate. In this review, these widely used quantum gates involving their equations, basic transformations, matrix representations, and notations are provided in Table 2.

(4) *Quantum circuits*

The modification of a quantum state can be described using the lexicon of quantum computation. A quantum circuit serves as a foundational model for quantum computation, where computations are conveyed through sequences of quantum gates (Nielsen and Chuang 2010). Note that the quantum circuit is interpreted from left to right, with each line corresponding to a wire within the quantum circuit. Although this representation may not directly correspond to a tangible physical wire, it may instead symbolize temporal progression, or even the behavior of physical particles. Conventionally, the state input to the circuit is presumed to be a computational basis state, typically representing a state composed entirely of $|0\rangle$s. In quantum circuits, the state of qubits undergoes natural evolution over time. Specific quantum algorithms are realized through various combinations of quantum gates, culminating in the presentation of results via quantum measurements.

In the domain of quantum computing, particularly within the framework of the quantum circuit model of computation, a quantum gate constitutes a fundamental quantum circuit

Table 2 Commonly used quantum gates including their notations, equations, and basic transformations

Gate	Equation	Matrix	Transform	Notation
Identity (**I**)	$\mathbf{I} = \|0\rangle\langle 0\| + \|1\rangle\langle 1\|$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\mathbf{I}\|0\rangle = \|0\rangle$ $\mathbf{I}\|1\rangle = \|1\rangle$	
Pauli-*X* (**X** or **NOT**)	$\mathbf{X} = \|0\rangle\langle 1\| + \|1\rangle\langle 0\|$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\mathbf{X}\|0\rangle = \|1\rangle$ $\mathbf{X}\|1\rangle = \|0\rangle$	
Hadamard (**H**)	$\mathbf{H} = \frac{\|0\rangle+\|1\rangle}{\sqrt{2}}\langle 0\| + \frac{\|0\rangle-\|1\rangle}{\sqrt{2}}\langle 1\|$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$\mathbf{H}\|0\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$ $\mathbf{H}\|1\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$	
Controlled-NOT (**CNOT**)	$\mathbf{CNOT} = \|0\rangle\langle 0\| \otimes \mathbf{I} + \|1\rangle\langle 1\| \otimes \mathbf{X}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	$\mathbf{CNOT}\|00\rangle = \|00\rangle$ $\mathbf{CNOT}\|01\rangle = \|01\rangle$ $\mathbf{CNOT}\|10\rangle = \|11\rangle$ $\mathbf{CNOT}\|11\rangle = \|10\rangle$	
Toffoli (**T** or **CCNOT**)	$\mathbf{T} = \|0\rangle\langle 0\| \otimes \mathbf{I} \otimes \mathbf{I} + \|1\rangle\langle 1\| \otimes \mathbf{CNOT}$	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$	$\mathbf{T}\|000\rangle = \|000\rangle$ $\mathbf{T}\|001\rangle = \|001\rangle$ $\mathbf{T}\|010\rangle = \|010\rangle$ $\mathbf{T}\|011\rangle = \|011\rangle$ $\mathbf{T}\|100\rangle = \|100\rangle$ $\mathbf{T}\|101\rangle = \|101\rangle$ $\mathbf{T}\|110\rangle = \|111\rangle$ $\mathbf{T}\|111\rangle = \|110\rangle$	

designed to manipulate a limited number of qubits. Quantum circuits commonly utilize various gates (see Table 2) to simplify complex transformations. A gate that operates on k qubits is typically represented as a $2^k \times 2^k$ unitary matrix. Given that quantum gates are expressed as unitary matrices, it follows that the number of qubits in both the input and output states of the gate must be identical. The replication of an unknown quantum state via a circuit is inherently impossible. This characteristic, known as the no-cloning theorem, illustrates the inability to replicate qubits, and is regarded as a fundamental distinction between quantum and classical information paradigms.

(5) *Quantum-enabled algorithms*

An algorithm is a step-by-step procedure or set of rules that a computer program or a computing device follows to perform a specific task or solve a problem. It is used to solve a wide variety of problems, from basic sorting and searching to complex optimization and machine learning tasks. In this review, the term “quantum-enabled algorithms" broadly refers to algorithms that either directly leverage the advantages of quantum states in their design, or indirectly draw upon quantum concepts and insights during the algorithm development process, with the common goal of enhancing the performance of existing algorithms. In this context, we thus categorize algorithms into two groups–quantum algorithms and quantum-inspired algorithms. Given that this field is still in the early stages of development, we encourage readers to refer to the open questions discussed in Sect. 5.1.2 when trying to understand these concepts and discuss their applications. Many quantum-enabled algorithms have been developed to address various challenges in multiple computer science fields, and have been employed in the applications and tasks of medical image processing. Several such algorithms are shown in Fig. 3, and typical techniques are discussed below, along with their specific applications in medical image processing. For reference, a comprehensive list of abbreviations for these algorithms is provided in the Appendix located at the end of this review.

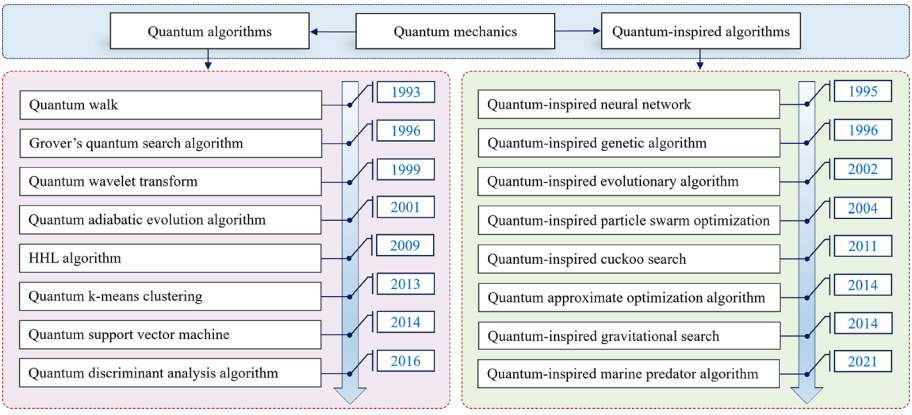


Fig. 3 An overview of notable quantum and quantum-inspired algorithms used in medical image processing. The year of the algorithm proposal is indicated

2.2 Quantum algorithms

Quantum algorithms are specifically developed using the principles of quantum mechanics to run on a realistic quantum device, which can solve specific problems more efficiently and faster than classical algorithms. Even though "realistic quantum devices" are mentioned in the definition, in the current landscape of computer science and related fields for practical applications, algorithms that involve the compilation, transformation, and measurement of quantum states as part of their steps are typically considered quantum algorithms, even if they are only simulated on a classical computer or as a specific module within a larger computational task. Some typical quantum algorithms widely used in medical image processing, such as quantum walk (QW), the Harrow–Hassidim–Lloyd (HHL) algorithm, and quantum clustering, are introduced as follows:

(1) *Quantum walk*

Kendon proposed a QW, also known as a quantum random walk, which involves discrete-time and continuous-time QW models (Kendon 2006). The coin and walker are the two quantum systems in a one-dimensional single-particle discrete-time QW model. The total QW system can be represented as a tensor product of the walker state (H_p) and the coin state (H_c) in the Hilbert space (H): $H = H_p \otimes H_c$. The evolution process of the entire quantum system can be expressed as follows:

$$U = S \cdot (I \otimes C), \tag{12}$$

where I indicates the identity operator, C represents the coin operator applied to the coin state, and S represents a conditional shift operator that can be represented as follows:

$$S = \sum_{x} \left(|x+1, 0\rangle\langle x, 0| + |x-1, 1\rangle\langle x, 1| \right). \tag{13}$$

Therefore, after walking step t, the final state $|\psi\rangle$ is represented as follows:

$$|\varphi\rangle_t = (U)^t |\varphi\rangle_{initial}, \tag{14}$$

where $|\varphi\rangle_{initial}$ denotes the initial state of the total quantum system. In addition, the probability that the walker is in position x is expressed as follows:

$$P(x, t) = \sum_{v \in 0,1} \left| \langle x, v | \varphi \rangle_t \right|^2. \tag{15}$$

As shown in Fig. 4, the distribution generated by an initial QW state demonstrates a nonlinear mapping characteristic, which can be thus regarded as an efficient generator of pseudo-random numbers. The highly randomized key generated through a pseudo-random number generator enhances the encryption strength for images, significantly augmenting the complexity of decryption processes. Consequently, many researchers have studied QW-based medical image encryption techniques to protect patient privacy and enhance security (Abd EL-Latif et al. 2020; Abd-El-Atty et al. 2023).

(2) *Harrow–Hassidim–Lloyd algorithm*

Linear systems of equations to be solved are fundamental tasks in many fields of mathematics, science, and engineering. Harrow et al. proposed a quantum algorithm for linear systems of equations, that is, the HHL algorithm (Harrow et al. 2009). In general, the HHL algorithm can be divided into three subroutines: phase estimation, control rotation, and inverse phase estimation. In particular, the circuit of the HHL algorithm is shown in Fig. 5.

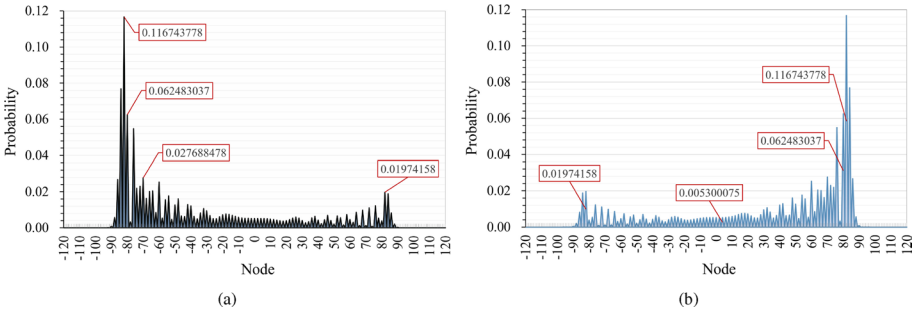


Fig. 4 Probability distribution generated for a quantum walk on a line (120 steps). The initial coin states in (a) and (b) are denoted by $|0\rangle$ and $|1\rangle$, respectively

The algorithm seeks to solve $A\vec{x} = \vec{b}$ using a quantum computer, where the Hermitian matrix A of $N \times N$ and vector $\vec{b}$ are the given conditions. First, $\vec{b}$ is represented as quantum state $|b\rangle = \sum_{i=1}^{N} b_i|i\rangle$. The next step is to decompose $|b\rangle$ into the eigenvector basis of A using phase estimation. The state of the entire system after the phase estimation subroutine is $\sum_{j=1}^{N} \beta_j|\mu_j\rangle|\lambda_j\rangle$, where μ_i denotes the eigenvector basis of A and λ_j denotes the corresponding eigenvalue. Then, a controlled rotation operation is performed, which is intended to convert $|\lambda_j\rangle$ to $\lambda_j^{-1}|\lambda_j\rangle$. The state of the entire system becomes $\sum_{j=1}^{N} \lambda_j^{-1}\beta_j|\mu_j\rangle|\lambda_j\rangle$. Finally, using inverse quantum phase estimation to uncompute the $|\lambda_j\rangle$ register, the algorithm yields a quantum state that is proportional to

$$\sum_{j=1}^{N} \lambda_j^{-1}\beta_j|\mu_j\rangle = A^{-1}|b\rangle = |x\rangle. \tag{16}$$

Compared with classical algorithms for solving linear equations, the HHL algorithm exhibits an exponential acceleration effect in certain cases and as a submodule. Therefore, it is widely used in some machine learning algorithms to form quantum machine learning algorithms, such as the quantum support vector machine (Rebentrost et al. 2014), quantum generative adversarial networks (Zoufal et al. 2019), and quantum principal component analysis (Lloyd et al. 2014). Quantum machine learning algorithms are utilized in various medical image processing tasks, notably in medical image classification (Wei et al. 2023), where they serve as proficient classifiers.

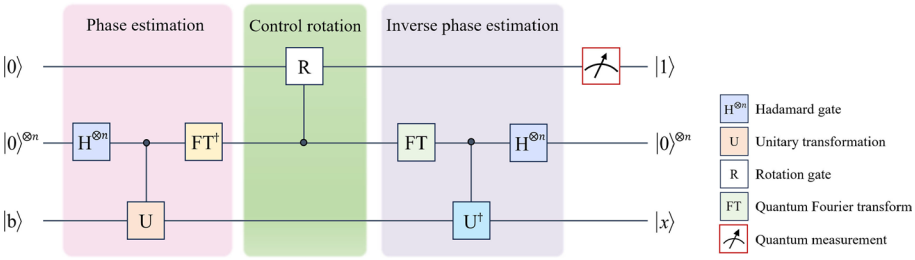


Fig. 5 Quantum circuit implementing the process of the HHL algorithm. "FT" denotes the quantum Fourier transform

(3) *Quantum k-means clustering*

A clustering algorithm is an essential and effective tool in data mining research and applications that separates data into many clusters based on similarities in data characteristics (Ezugwu et al. 2022). To overcome the challenges of computational speed and efficiency for large datasets, a quantum k-means clustering algorithm that incorporates principles of quantum mechanics was proposed by Lloyd et al. in 2013 (Lloyd et al. 2013).

In the k-means clustering algorithm, classical data ($\vec{\mu} = \vec{\mu}_1, \vec{\mu}_2, \ldots, \vec{\mu}_n$) need to be transformed into quantum data, that is, $|\mu\rangle$. The state of the system at the beginning of the protocol is defined as $|0, \phi, \psi\rangle$. Herein, two ancilla quantum states are defined:

$$|\psi\rangle = (1/\sqrt{2})(|0\rangle|\mu\rangle + (1/\sqrt{M})\sum_{j=1}^{M}|j\rangle|v_j\rangle), \tag{17}$$

$$|\phi\rangle = (1/\sqrt{Z})(|\vec{\mu}||0\rangle - (1/\sqrt{M})\sum_{j=1}^{M}|\vec{v_j}||j\rangle), \tag{18}$$

where $|\vec{\mu}|$ denotes the norm of vector $\vec{\mu}$, M is the number of the samples, $\vec{v_j}$ indicates the jth vector in a cluster, and $Z = |\vec{\mu}|^2 + (1/M)\sum_j |\vec{v}_j|^2$. After the first Hadamard gate, the state of the system becomes $1/\sqrt{2}(|0, \phi, \psi\rangle + |1, \phi, \psi\rangle)$. A control-swap gate is then used to evaluate the similarity between quantum states. Finally, the second Hadamard gate results in

$$\frac{1}{\sqrt{2}}|0\rangle(|\phi\rangle|\varphi\rangle + |\varphi\rangle|\phi\rangle) + \frac{1}{\sqrt{2}}|1\rangle(|\phi\rangle|\varphi\rangle - |\varphi\rangle|\phi\rangle). \tag{19}$$

The measurement gate on the first qubit ensures that it is 0 with a probability of

$$P(|0\rangle) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}|\langle\phi, \varphi\rangle|^2. \tag{20}$$

The random selection of the initial cluster center causes the algorithm to fall into a local optimal solution to a great extent. To solve this problem, Lloyd et al. presented a method for a quantum adiabatic algorithm to find initial seeds for clusters (Lloyd et al. 2013). In addition, many improved quantum clustering algorithms have recently been proposed (Kerenidis et al. 2019; Hou et al. 2022; Wu et al. 2022). The quantum clustering algorithm exhibits exponential speedup over classical algorithms for large datasets. Similar to traditional clustering algorithms, quantum clustering algorithms play a significant role in extracting salient features, segmenting image regions, and identifying pathological areas. In recent years, quantum clustering has been used to solve various problems in the medical field, such as CT scan image segmentation (Singh and Bose 2021), heart disease detection (Kavitha and Kaulgud 2023), and medical data classification (Deshmukh et al. 2023).

2.3 Quantum-inspired algorithms

Quantum-inspired algorithms are designed based on certain principles of quantum mechanics to operate on classical devices with the aim of achieving and exceeding the performance of previously known classical methods. In contrast to the earlier described quantum algorithms,

quantum-inspired algorithms generally do not involve the actual transformation of quantum states. This statement may be imprecise, even if we were to confine our discussion to specific applications within the field of computer science, such as medical image processing, as there are some exceptions in the existing literature. This review aims to comprehensively and clearly convey the current development of medical image processing techniques that leverage quantum-enabled algorithms, while respecting the terminology used by the original authors in their descriptions of the relevant algorithms. For further discussion, please refer to the open questions presented in Sect. 5.1.2. Several quantum-inspired algorithms widely employed in the medical field, that is, quantum-inspired particle swarm optimization (QPSO), quantum-inspired genetic algorithm (QGA), and quantum-inspired neural network (QINN), are presented as follows.

(1) *Quantum-inspired particle swarm optimization algorithm*

The QPSO algorithm is quantum mechanics applied to the traditional particle swarm optimization (PSO) algorithm, improving its exploration and exploitation capabilities (Sun et al. 2004). In the PSO algorithm, a particle is characterized by its position vector $\vec{x}$ and velocity vector $\vec{v}$ to determine the trajectory of the particle. However, the term trajectory is meaningless in the QPSO algorithm because $\vec{x}$ and $\vec{v}$ of a particle cannot be determined simultaneously according to the uncertainty principle of quantum mechanics.

Each particle in the background of a quantum system can be considered a quantum state, denoted by a wave function $\theta(\vec{x}, t)$. In three-dimensional space, the wave function can be represented as follows:

$$|\theta|^2 dxdydz = Fdxdydz, \tag{21}$$

where $|\theta|^2$ denotes the probability density function, which satisfies

$$\int_{-\infty}^{+\infty} |\theta|^2 dxdydz = \int_{-\infty}^{+\infty} Fdxdydz = 1. \tag{22}$$

In the QPSO algorithm, all parameters are first initialized. To obtain the position of the particle, the Monte Carlo method was used to simulate the measurement process. The particles move according to the following equations:

$$X_i(t+1) = \begin{cases} p_i(t) + \alpha \cdot |m_{i,best} - X_i(t)| \cdot ln\frac{1}{\mu_i(t)}, & if\ s \geq 0.5 \\ p_i(t) - \alpha \cdot |m_{i,best} - X_i(t)| \cdot ln\frac{1}{\mu_i(t)}, & Otherwise \end{cases}, \tag{23}$$

where u_i and s denote uniformly distributed random numbers in the interval [0,1], and α denotes the contraction-expansion coefficient. In addition,

$$p_i(t) = \varphi_i(t) \cdot P_{i,best}(t) + (1 - \varphi_i(t)) \cdot P_{global}(t), \tag{24}$$

$$m_{i,best} = \frac{1}{Q} \sum_{i=q}^{Q} P_{i,best}(t), \tag{25}$$

where φ_i denotes a uniformly distributed random number in the interval [0,1], Q represents the number of all particles, and $m_{i,best}$ is defined as the mean of the best positions of all particles in the population. If the termination criteria are met, return the best solution found, that is, the particle with the best fitness value.

The PSO algorithm can yield superior solutions to complex problems, making it widely applicable in medical image processing tasks (Esmin et al. 2015; Tan et al. 2019). The QPSO algorithm also exemplifies outstanding performance in optimization problems, and therefore can be effectively employed to sort out various complex problems in the medical field, such as medical image segmentation (Radha and Gopalakrishnan 2020) and feature selection (Shi et al. 2019) in disease diagnosis.

(2) *Quantum-inspired genetic algorithm*

QGA was proposed by Narayanan and Moore, in which the principles of quantum mechanics were used to inspire a more efficient evolutionary computing method for the traveling salesperson problem (Narayanan and Moore 1996). Note that QGA is developed based on the basic skeleton of the genetic algorithm (GA), resulting in many similar steps in the algorithm (Dahi et al. 2016).

A chromosome in the QGA can be represented by the following qubit:

$$q_j^t = \left[\begin{array}{c|c|c|c} \alpha_1 & \alpha_2 & \cdots & \alpha_m \\ \beta_1 & \beta_2 & \cdots & \beta_m \end{array}\right], \tag{26}$$

where m denotes the number of qubits and $j \in [1, n]$, with n denoting the size of the population.

The qubit chromosome is updated by applying some appropriate quantum gates, such as a rotation gate, that is,

$$R(\Delta\theta_j) = \begin{bmatrix} cos(\Delta\theta_j) & -sin(\Delta\theta_j) \\ sin(\Delta\theta_j) & cos(\Delta\theta_j) \end{bmatrix}, \tag{27}$$

where $\Delta\theta_j$ denotes the rotation angle. In particular, the updated i-th qubit $[\alpha_i',\ \beta_i']^T$ can be obtained using the following equation:

$$[\alpha_i',\ \beta_i']^T = R(\Delta\theta_j) \cdot [\alpha_i,\ \beta_i]^T. \tag{28}$$

It is encoded as a probability vector with real values via quantum measurement. In binary optimization problems, the following equation allows for the implementation of quantum measurement:

$$X = \begin{cases} 1, & \text{if } rand \le |\alpha|^2, \\ 0, & \text{Otherwise}, \end{cases} \tag{29}$$

where *rand* denotes a random number. Notably, if the termination criteria are satisfied, the best solution can be obtained.

The performance of QGA designed by the principles of quantum mechanics in solving complex problems is superior to that of traditional GA (Abdulhussien et al. 2023). Recently, QGA has been widely used in the medical field such as in medical image segmentation (Das et al. 2019, 2020; Hilali-Jaghdam et al. 2020). In these instances, the QGA was employed to optimize the parameters of various models, with the aim of achieving the best results.

3) *Quantum-inspired neural network*

QINNs are based on the combination of quantum computing with the properties of neural networks that can be implemented on a classical computer. Kak first proposed the concept of quantum neural computing (Kak 1995), which has attracted the interest of many researchers. Subsequently, a QINN was first designed by Menneerf and Narayanan in 1995,

demonstrating the practical benefits of adopting quantum concepts in neural networks (Menneer and Narayanan 1995). To date, many versions of QINNs have been developed. Most proposals for QINN models are based on the qubit neuron model, in which the neuron states are connected to the quantum states, and the transitions between neuron states are based on operations derived from quantum logic gates.

A schematic diagram of the QINNs used to implement medical image segmentation is shown in Fig. 6. The network generally comprises three layers: input, hidden, and output layers (Ganjefar and Tofighi 2018). Herein, the input layer is used to convert the input data into quantum states. The hidden layer is the intermediate layer in a QINN, with neurons that receive information from the input layer and process it. The output layer receives the processed information from the hidden layer and converts it into the final output.

Analogous to traditional convolutional neural networks, QINNs can be employed to address classification challenges. In recent years, a great number of studies based on QINNs have been proposed to address various complex problems, such as text classification (Shi et al. 2023), emotion recognition (Li et al. 2021), and disease diagnosis (Parisi et al. 2022).

3 Disease diagnosis in quantum-assisted medical image processing

Early accurate diagnosis and treatment of diseases are paramount in reducing premature mortality rates (Yan et al. 2024). The process of disease diagnosis based on medical images can generally be divided into four main stages: image preprocessing, segmentation of the region of interest (ROI), feature extraction or selection, and classification (Hasoon et al. 2021; Yan et al. 2023). Although classical machine learning has demonstrated significant potential in disease diagnosis tasks, it faces challenges due to the inherently insufficient labeled medical data, which leads to suboptimal processing efficiency and diagnostic accuracy. In contrast, quantum machine learning has the potential to achieve remarkable advancements in parameter optimization, execution efficiency, and error rate reduction in medical image processing tasks, by leveraging the superposition and entanglement properties of quantum states (Wei et al. 2023). This section dwells on recent work on image preprocessing, ROI segmentation, and classification in disease diagnosis using quantum-enabled algorithms. Notably, the utilization of quantum methods in the domain of feature extraction/selection techniques has received relatively limited attention in the past years. Here, the framework for disease diagnosis based on medical images using quantum-enabled algorithms is depicted in Fig. 7.

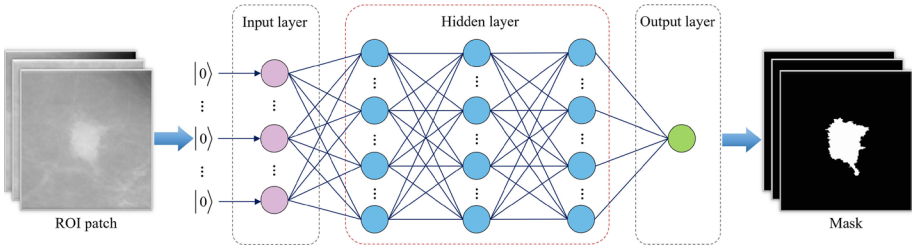


Fig. 6 A diagram of quantum-inspired neural network for medical image segmentation

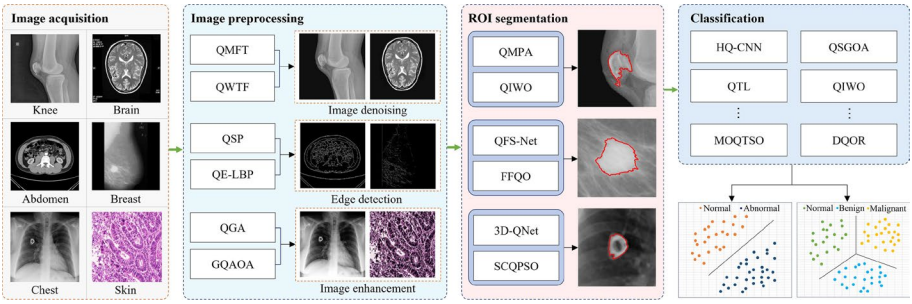


Fig. 7 A framework of disease diagnosis in quantum-assisted medical image processing includes the following main stages: image preprocessing, ROI segmentation, and image classification

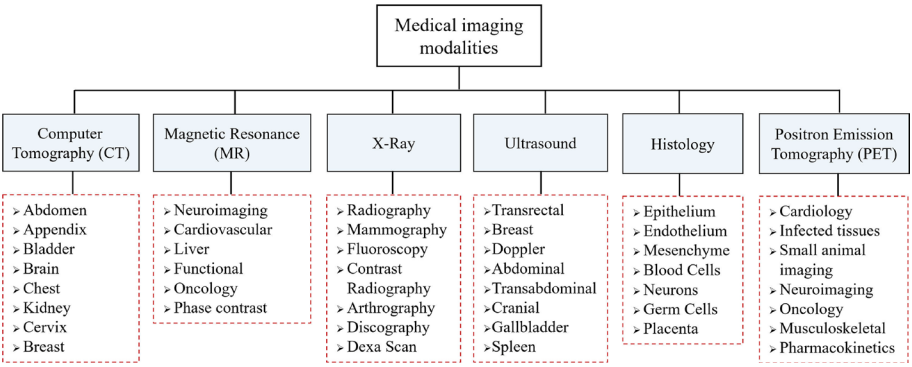


Fig. 8 Modalities of medical imaging and applications in diagnosing diseases in various parts of the human body

3.1 Preprocessing of medical image

As shown in Fig. 8, many different medical imaging technologies can be used to generate medical images of organs of the human body to perform a specific disease diagnosis. Typically, medical images acquired using various imaging technologies inevitably incorporate noise and trivial objects, which significantly reduce the overall image quality and adversely affect the accuracy and reliability of disease diagnosis results. In particular, the preliminary processing of medical images constitutes a pivotal stage in the construction of precise machine learning models dedicated to disease diagnosis (Xu et al. 2019). Therefore, it is essential to remove these noises and insignificant objects from medical images to further improve the accuracy of disease detection and diagnosis. In this subsection, related studies employing quantum-enabled algorithms for preprocessing medical images are summarized.

In the early stages, Fu et al. proposed a new medical image despeckling method based on quantum-inspired bilateral filtering (QIBF) and wavelet thresholding (Fu et al. 2015). In this process, QIBF was utilized as a preprocessing step to improve the despeckling performance. The experimental results demonstrate that the proposed method can significantly reduce speckle noise and retain high-contrast features. However, this method lacks the adaptability to determine the parameters of the statistical models for the wavelet

coefficients of different images. To solve these issues, Fu et al. proposed a novel adaptive image despeckling algorithm to improve the quality of medical images (Fu et al. 2015). In this scheme, a quantum-inspired thresholding function (QITF) based on the characteristics of the coefficients in dual-tree complex wavelet transform (DTCWT) was proposed to be incorporated into a Bayesian framework. To verify the effectiveness of the proposed scheme, several experiments were performed, and the experimental results indicated that the scheme can suppress speckle noise and preserve the details of medical images with relatively low computational complexity. Moreover, Ahmadi et al. introduced an advanced noise removal method that utilizes a combination of local thresholding, active contour, and an optimized quantum matched-filter technique (QMFT) (Ahmadi et al. 2021). In this framework, the QMFT functioned as a local search operator, enhancing the initial images to minimize noise in the magnetic resonance (MR) images. Similarly, Qasim et al. introduced a methodology that integrates local thresholding, active contour, and a quantum wavelet transform filtering (QWTF) approach to effectively eliminate noise in mammographic images (Qasim et al. 2022).

Medical images frequently exhibit intricate characteristics, including dark backgrounds, indistinct boundaries, and low contrast (Zhou et al. 2019; Huang et al. 2020). Edge detection technology can be utilized to maintain more explicit edge details in medical images to highlight important information regarding the size and boundary location of the nidus or tumors in medical images. Thus, Widiyanto et al. proposed an edge detection scheme for medical images using an improved quantum-Canny (IQC) algorithm (Widiyanto et al. 2019). In this scheme, the edge detection outcome is enhanced by computing the gradient magnitude value based on the neighboring pixels that represent the strength of the center pixel. The proposed method can provide higher peak signal-to-noise ratio (PSNR) and lower mean square error (MSE) values than the original Canny, Sobel, Robert, and Prewitt operators. Note that texture features play a pivotal role in the discrimination of diverse tissues in medical images. Lekehali et al. proposed a quantum local binary pattern (QuLBP) to consider texture information to effectively detect edges in MR images (Lekehali and Moussaoui 2019). In addition, Jamal et al. proposed a novel breast cancer edge detection method for mammography images based on QGA and a support vector machine (Tariq Jamal et al. 2021). In this method, QGA was used to solve a multilevel thresholding problem based on the Tsallis entropy. The proposed approach was compared to some standard methods of edge detection in a sample of mammographic images taken from a well-known benchmark database, which demonstrated the effectiveness of the proposed approach. Note that edge detection can effectively improve the classification accuracy of medical images. For example, Rubio et al. introduced a novel methodology based on quantum signal processing (QSP) and cellular automata to detect microcalcifications in mammograms (Rubio et al. 2019); additionally, Hasan et al. proposed a brain MR image classification scheme using quantum entropy-local binary pattern (QE-LBP) and deep-learning features (Hasan et al. 2020).

In order to address the issues of poor quality and low contrast in medical images, image enhancement technology has been developed and has become an increasingly significant preprocessing step in the diagnosis of medical image-based diseases. For example, in an early-stage study by Wang et al., a novel optimal scheme was proposed to enhance retinal images using a combination of the curvelet transform and QGA (Wang et al. 2013). In this scheme, the curvelet coefficients were modified using a nonlinear enhancement mapping function. The QGA was then used to select optimal parameters for the nonlinear mapping function. Recently, Pashaei et al. proposed a medical image enhancement scheme using Gaussian quantum arithmetic optimization (GQAOA)-based histogram equalization and

contrast-limited adaptive histogram equalization (CLAHE) to improve the content of the information and details of medical images (Pashaei and Pashaei 2023). In this process, the GQAOA was used to automatically find the optimal values to tune CLAHE's parameters. Finally, a comparative study was conducted to demonstrate the superiority of the proposed image enhancement scheme by comparing it with state-of-the-art techniques.

In summary, the preprocessing step for medical images, including noise removal, edge detection, and image enhancement, was designed based on various quantum-enabled algorithms. In addition, a comprehensive description of the previous studies is presented in Table 3, providing detailed information on each work. Table 3 presents the average evaluation metrics of the experimental results obtained from these studies, demonstrating their effectiveness over the traditional methods. It is noteworthy that the results of the various metrics for each scheme are presented in the table, primarily for intuitive reference rather than technical comparison, as the cited studies may have employed distinct experimental settings.

3.2 ROI segmentation in medical image

Medical image segmentation is an essential assignment in computer-aided diagnosis and computer-aided surgery (Yao et al. 2022). Generally, medical image segmentation can be divided into traditional methods relying on hand-crafted features and deep learning technique-based segmentation methods. Automated and accurate segmentation of lesions in medical images can provide effective and reliable diagnostic evidence for radiologists, helping to quantitatively evaluate disease prognosis and treatment efficacy (Liu et al. 2020; Lin et al. 2022). In this section, related studies employing quantum-enabled algorithms for medical image segmentation are outlined.

Threshold-based segmentation is one of the most widely used techniques because it is simple and easy to implement (Houssein et al. 2021; Nama 2022). Therefore, many researchers are dedicated to the study that applied the method of combining threshold-based segmentation and quantum algorithms for medical images to select optimal ROI for disease diagnosis. For example, Li et al. proposed a partitioned and cooperative quantum particle swarm optimization (SCQPSO) algorithm that was applied to medical image segmentation using a multilevel thresholding method (Li et al. 2017). In this scheme, the SCQPSO algorithm was used to optimize the parameters of the Otsu thresholding method to obtain the maximum value of the between-class variance and achieve high accuracy in for medical image segmentation. Subsequently, Hilali–Jaghdam et al. proposed multilevel segmentation methods of medical images based on the GA and QGA (Hilali-Jaghdam et al. 2020). The two algorithms are employed to maximize the Rényi, Masi, and Shannon entropies, aiming to achieve the exact segmentation of medical images. Twenty medical images were obtained to evaluate the effectiveness of the proposed scheme. The results indicated that QGA outperformed GA in segmented tasks, and the Rényi entropy was superior to other methods for medical image multilevel thresholding. Consequently, many medical image segmentation schemes have been proposed based on a combination of multilevel thresholding and quantum-inspired optimization algorithms. For example, the quantum-inspired marine predator algorithm (QMPA) and quantum-inspired firefly algorithm (QIFA) were used to substantially improve the quality of medical image segmentation (Abd Elaziz et al. 2021; Choudhury et al. 2022).

In addition, the cluster-based segmentation method is trendy and practical and can retain more image details (Oskouei et al. 2021). Quantum-enabled algorithms can be employed

Table 3 Quantum-enabled algorithms for preliminary processing of medical images

Applications	Authors	Year	Quantum algorithms	Other methods	Image modalities	Metrics
Denoising	Fu et al. (2015)	2015	QIBF	DTCWT	Ultrasound	SNR[a] = 16.75 SSIM[b] = 0.88
	Fu et al. (2015)	2015	QITF	DTCWT	Ultrasound	SNR = 12.13 ENL[c] = 27.87
	Ahmadi et al. (2021)	2021	QMFT	Local thresholding	MR	PSNR = 87.35 MSE = 0.006
	Qasim et al. (2022)	2022	QWTF	Active contour	Mammography	PSNR = 38.83 MSE = 0.70
Edge detection	Widiyanto et al. (2019)	2019	IQC	Gradient magnitude	CT	PSNR = 66.45 MSE = 0.015
					X-ray	PSNR = 66.58 MSE = 0.017
	Rubio et al. (2019)	2019	QSP	Cellular automaton	Mammography	AUC1 = 0.881 AUC2 = 0.859
	Lekehali and Moussaoui (2019)	2019	QuLBP	Cellular automaton	MR	SSIM = 0.591
	Hasan et al. (2020)	2020	QE-LBP	LSTM[f] network	MR	Precision = 0.970 Accuracy = 0.988
	Tariq Jamal et al. (2021)	2021	QGA	Multilevel thresholding	Mammography	PSNR = 7.97 SSIM = 0.301 FSIM[d] = 0.729
Image enhancement	Wang et al. (2013)	2013	GQA	Curvelet transform	Retinal image	RMSE = 20.53 PSNR = 21.88
	Pashaei and Pashaei (2023)	2023	GQAOA	CLAHE	MR	SSIM = 0.844 PSNR = 22.67 NIQE[e] = 3.398

[a]SNR: Signal-to-Noise Ratio, [b]SSIM: Structural Similarity Index Measure, [c]ENL: Equivalent Number of Looks, [d]FSIM: Feature Similarity Index, [e]NIQE: Naturalness Image Quality Evaluator, [f]LSTM: Long Short-term Memory

to optimize various clustering techniques, with the aim of improving the performance of cluster-based medical image segmentation. Das et al. presented a color MR image segmentation method based on fuzzy c-means clustering and a quantum-inspired modified genetic algorithm (QIMGA) (Das et al. 2019). In this process, QIMGA was used to address the issues of early convergence to local minima in fuzzy c-means clustering to further mitigate the associated challenges of cost and time in segmentation tasks. Similarly, Singh et al. proposed a novel medical image segmentation method for chest CT images based on a k-means clustering algorithm and a fast forward quantum optimization (FFQO) algorithm (Singh and Bose 2021). The k-means clustering algorithm was used to cluster the gray-level values of the images, and the FFQO algorithm was used to obtain the optimal cluster centers and global optimum. The effectiveness of the proposed method was verified on chest CT images of coronavirus 2019 (COVID-19) patients using the evaluation metrics of MSE, PSNR, Jaccard similarity coefficient (JSC), and correlation coefficient (CC), which demonstrated the superiority of the proposed method over some traditional and nontraditional methods in terms of segmented results.

Segmentation methods based on deep learning are highly effective and accurate for image segmentation (Minaee et al. 2021; Soomro et al. 2022; Zhou et al. 2024). First, quantum algorithms can be employed to optimize the parameters of the neural network models, thereby leading to improved segmentation results. Razmjooy et al. proposed a skin melanoma segmentation method using neural networks and quantum invasive weed optimization (QIWO) algorithm (Razmjooy and Razmjooy 2020). The QIWO algorithm was utilized to optimize the weights of the neural network to avoid trapping into the local minimum, which aimed to achieve the exact separation of the skin lesion regions in dermatological images. DermIS and Dermquest databases were used to analyze and verify the efficiency of the proposed scheme for the segmentation of the skin lesion. Furthermore, Ahmadi et al. proposed a medical image segmentation scheme using a deep spiking neural network with the quantum artificial immune system (QAIS) to segment the tumor area of MR images (Ahmadi et al. 2021). In this scheme, QAIS was used to optimize the segmentation process during network training, and the simulation results using the BraTS 2018 dataset demonstrated a notable advantage over similar previous methods.

Second, a quantum neural network (QNN) is a new computing model that combines quantum computing and neural network theory, and it can be applied to medical image segmentation to provide more efficient and accurate segmented results. For example, Konar et al. proposed a quantum-inspired self-supervised neural network (QIS-Net) for automated brain MR image segmentation (Konar et al. 2020). The QIS-Net model used a self-supervised procedure guided by the counterpropagation of the network states to obviate the complex quantum backpropagation algorithms employed in supervised QINN models. The effectiveness of the scheme in detecting tumors was validated using dynamic susceptibility contrast brain MR images from the Nature datasets, and the results yielded good accuracy and reasonable dice similarity (DS) scores. Subsequently, Konar et al. again proposed a novel self-supervised shallow learning network model exploiting the sophisticated three-level qutrit-inspired quantum information system (i.e., quantum fully self-supervised neural network, QFS-Net) for automated brain MR image segmentation (Konar et al. 2021). The pixel intensities and interconnection weight matrix were expressed in quantum formalism in classical simulations, thereby reducing the computational overhead and enabling faster convergence of the network states. The suggested model was validated using the Cancer Imaging Archive (TCIA) dataset, where the results indicated promising segmented outcomes in tumor detection in terms of DS and accuracy with minimal human intervention and computational resources. To protect the patient's personal information,

Amin et al. proposed a brain tumor segmentation method based on lightweight quantum and convolutional neural networks (LQCNNs) (Amin et al. 2022). In this scheme, MR images were encrypted and decrypted using the SHA-256 algorithm to enhance their security. The decrypted images were supplied to the 2-qubit quantum model for tumor classification; then, the classified tumor images were input into a segmentation model for precise tumor segmentation. The proposed model on the BraTS 2020 dataset can obtain a 98% DS coefficient. Additionally, in 3D medical image segmentation tasks, 3D deep convolutional neural networks (DCNNs) have demonstrated striking improvements over their 2D counterparts, as they can explore contextual information contained in slices that can contribute significantly to better segmentation performance (Zhang et al. 2020). Thus, Konar et al. proposed a novel shallow 3D self-supervised tensor neural network in quantum formalism (i.e., three-dimensional quantitative network (3D-QNet)) for volumetric segmentation of medical images with the merits of obviating training and supervision (Konar et al. 2023). The incorporation of tensor decomposition in quantum formalism led to faster convergence of network operations to preclude the inherent slow convergence problems faced by classical supervised and self-supervised networks. The proposed scheme was evaluated using the BraTS 2019 dataset and the Liver Tumor Segmentation Challenge (LiTS17) dataset to demonstrate the potential advantage of the proposed network in medical image segmentation tasks.

As shown in Table 4, we summarized the previous relevant studies on medical image segmentation using quantum-enabled algorithms. Furthermore, some relevant studies are included in the table but are not covered in this review. To facilitate a comprehensive understanding of the recent advances in medical image segmentation, we further outlined the image modalities, organs, and data sources used in these studies. Note that the results of the evaluation metrics for each scheme were enumerated to reveal the effectiveness of quantum algorithms in medical image segmentation. Similarly, the results of various metrics for each scheme are displayed in the table mainly for intuitive reference rather than for direct technical comparison.

3.3 Medical image classification

The accurate classification of medical images can effectively help physicians detect and diagnose specific diseases. Quantum algorithms, notably quantum machine learning, have tremendous promise as tools that can handle the challenges of limited labeled medical data and high algorithm complexity in conventional machine learning. In this section, we recapitulate related studies in recent years that have employed quantum algorithms, particularly quantum machine learning, for the classification of medical images.

High-dimensional data can lead to poor performance and high computational costs for data classification. The feature selection technique can be used to eliminate redundant or irrelevant features from high-dimensional data to improve overall classification performance (Paniri et al. 2020; Song et al. 2021; Sun et al. 2021). Thus, Iliyasu et al. proposed a quantum hybrid intelligent approach for the detection of cervical cancer in cervical smeared images (Iliyasu and Fatichah 2017). In this process, a QPSO algorithm with an adaptive search capability was employed to select the best subset features to improve the classification accuracy of cervical cells for the detection and diagnosis of cervical cancer. Similarly, Shi et al. introduced a strategy that combines QPSO with random forest to predict the progression of idiopathic pulmonary fibrosis using high-resolution computed tomography (HRCT) images (Shi et al. 2019). In this scheme, QPSO was used to select the

Table 4 Quantum-enabled algorithms for medical images segmentation (the data sources are expounded in Appendix)

Authors	Year	Organs	Image modalities	Data sources	Methods	Metrics
Li et al. (2017)	2017	Stomach	CT	N/A	SCQPSO	St.Var[a] = 8.8307E-13
Das et al. (2019)	2019	Brain	MR	N/A	QIMGA	MR image1: Variance = 8.64E+05 MR image2: Variance = 8E+17
Hilali-Jaghdam et al. (2020)	2020	Brain/eye	N/A	N/A	QGA	PSNR = 25.71 SSIM = 0.775 FSIM = 0.879
Radha and Gopalakrishnan (2020)	2020	Brain	MR	N/A	IQPSO	DSC[b] = 0.887
Konar et al. (2020)	2020	Brain	MR	Nature dataset	QIS-Net	DSC = 0.748 Accuracy = 0.987 Sensitivity = 0.916
Razmjooy and Razmjooy (2020)	2020	Skin	Microscopic	DermIS/dermquest databases	QIWO	CDR[c] = 0.87 FAR[d] = 0.085 FRR[e] = 0.045
Abd Elaziz et al. (2021)	2021	Brain	MR	Brainweb database	QMPA	PSNR = 29.09 SSIM = 0.9164
Singh and Bose (2021)	2021	Chest	CT	COVID-19 datasets	FFQO	PSNR = 19.61 JSC = 0.90 CC = 0.91
Konar et al. (2021)	2021	Brain	MR	Nature dataset	QFS-Net	DSC = 0.785 PPV = 0.694 Accuracy = 0.988
Amin et al. (2022)	2022	Brain	MR	BraTS 2020	LQCNN	DSC = 0.985
Choudhury et al. (2022)	2022	Hippocampus	Microscopic	N/A	QIFA	PSNR = 31.17 SSIM = 0.949 Accuracy = 0.949
Prasad et al. (2023)	2023	Brain	MR	LGG Segmentation	3T-QCNN	Accuracy = 0.98 Sensitivity = 0.95 Specificity = 0.99
Liu et al. (2023)	2023	Brain	MR	ATLAS R2.0	SQMLP-Net	DSC = 0.709
Konar et al. (2023)	2023	Brain	MR	BraTS 2019	3D-QNet	DSC = 0.958 PPV[f] = 0.801 Accuracy = 0.989

[a]St.Var: Standard deviation; [b]DSC: Dice Similarity Coefficient; [c]CDR: Correct Detection Rate; [d]FAR: False Acceptance Rate; [e]FRR: False Rejection Rate; [f]PPV: Positive Predictive Value

optimal features to achieve high accuracy while maintaining a balance between sensitivity and specificity. Sharma et al. proposed a quantum-inspired binary bat algorithm (QIBBA) for the classification of leukocytes in blood smears (Sharma and Sharma 2022). QIBBA was used to eliminate irrelevant features extracted from leukocyte images to improve efficiency and reduce computational costs. On the other side, quantum algorithms can also be employed to optimize the parameters of neural networks utilized in medical image classification tasks. For example, Al-Wesabi et al. designed a deep learning model based on a multi-objective quantum tunicate swarm optimization (MOQTSO) algorithm for dystrophinopathies diagnosis (Al-Wesabi et al. 2023). The MOQTSO algorithm was used to optimize the initial seed points and thresholds of the region growth approach and tune the parameters of the extreme learning machine. The experimental results unequivocally indicate that the proposed method improves classification accuracy and represents a significant improvement in overall classification effectiveness.

As discussed earlier, quantum machine learning has the potential to accelerate data analysis, which has motivated numerous researchers to explore its application in medical image classification. For example, Amin et al. presented a brain tumor detection method based on a transfer learning technique and the quantum variational classifier (QVC) (Amin et al. 2022). In this study, the score vector was obtained from the pretrained Inception v3 model and then supplied to a six-layer quantum network model to predict the brain MR image in glioma, meningioma, pituitary tumor, and without tumor. Similarly, Toledo-Cortés et al. proposed a deep quantum ordinal regressor (DQOR) model for the diagnosis of medical images to serve as an effective diagnostic support tool for medical specialists (Toledo-Cortés et al. 2022). DQOR is a probabilistic model that combines a convolutional neural network (CNN) with quantum measurement regression. The performance of this model was validated on two medical image diagnosis tasks (i.e., diabetic retinopathy and prostate cancer diagnosis), and the experimental results indicated that it showed improved diagnostic accuracy compared to other methods. In addition, Landman et al. introduced two quantum methods for neural networks, quantum-assisted neural networks (QANN) and quantum orthogonal neural networks (QONN), applied to the classification of medical images (Landman et al. 2022). The scheme suggested that the proposed quantum networks generated a similar level of accuracy and demonstrated competitive scalability, thereby supporting the promise that quantum methods can be beneficial in solving visual tasks. Data-driven models with high performance are often based on more complex algorithms but lack explainability. As such, Deshmukh et al. introduced an explainable quantum clustering method to model medical data (Deshmukh et al. 2023). In this process, to diagnose abnormal activities based on breast cancer images and Knee MR image datasets to generate an explanation of the predictions, the proposed model used learning strategies, such as the local interpretable model-agnostic explanation method and the improved quantum k-means (Iqk-means) algorithm.

A hybrid classical-quantum neural network (CQNN) has remarkable speed-up potential for immense medical image datasets and can improve the performance of medical image classification. For example, Shahwar et al. proposed a hybrid CQNN model to automate the detection of Alzheimer's disease in brain MR images (Shahwar et al. 2022). In the CQNN model, a quantum variational circuit was used to generate feature vectors for precise decision boundaries. The CQNN model yielded a training accuracy of 99.1% and a test accuracy of 97.2%, which significantly outperformed the classical network. In addition, Ovalle-Magallanes et al. proposed a hybrid classical-quantum convolutional neural network (CQCNN) to classify stenosis and nonstenosis cases in X-ray coronary angiography images (Ovalle-Magallanes et al. 2022). Azevedo et al. explored a hybrid CQNN using

transfer learning to classify full-image mammograms into malignant and benign (Azevedo et al. 2022). Similarly, Choudhuri et al. proposed quantum-classical convolutional neural networks (QCCNNs) for brain MR image classification (Choudhuri and Halder 2023). The proposed QCCNN achieved accuracies in the range of 97.5%-–98.72% on different datasets. Obviously, the hybrid CQNN exhibits excellent accuracy performance in medical image classification tasks.

In recent years, COVID-19 has rapidly become an infectious disease with a high mortality rate, leading to numerous deaths. To reduce the death rate of the disease, quantum machine learning algorithms have been widely used to classify medical images to accurately diagnose and detect COVID-19. For example, to classify a positive or negative case of COVID-19 on lung CT images, Acar et al. presented a disease detection method using quantum transfer learning (QTL) (Acar and Yilmaz 2021). Similarly, Umer et al. proposed an integrated framework based on classical and QTL for COVID-19 classification on chest X-ray images (Umer et al. 2022). The performance of the proposed schemes in the COVID-19 classification was validated using various datasets, in which all experimental results consistently demonstrated the superiority of QTL over classical transfer learning. In addition, Sengupta et al. proposed a QNN model to determine whether patients had a COVID-19 infection (Sengupta and Srivastava 2021). The prediction accuracy and training time of the proposed QNN model were better than those of the conventional CNN model for medical image classification tasks. Similarly, Houssein et al. proposed a hybrid QCCNN model using random quantum circuits to detect COVID-19 patients in chest X-ray images (Houssein et al. 2022). Amin et al. investigated a quantum convolutional neural network (QCNN) for COVID-19 classification based on synthetic data generation (Amin et al. 2022). Notably, the performance of CNNs in medical image classification tasks can be significantly enhanced by improving their activation functions. Consequently, Parisi et al. introduced quantum rectified linear unit (QReLU) and modified QReLU (m-QReLU) activations for CNN to detect COVID-19 and Parkinson's disease diagnosis (Parisi et al. 2022). The effectiveness of the two novel activation functions proposed by the authors was validated using different datasets, producing highly accurate classification results.

As presented in Table 5, we further summarized the aforementioned related studies and some others on medical image classification using quantum algorithms (particularly quantum machine learning). The image modalities, data sources, and quantum-related algorithms employed for the classification of medical images were presented. Significantly, the experimental results of these studies using various metrics have been displayed to exemplify the advantages of quantum algorithms in medical image classification. Likewise, the table displays the results of various metrics for each scheme, primarily for intuitive reference rather than direct technical comparison.

4 Quantum-enabled algorithms for medical image security

Ensuring the secure storage and transmission of medical images is crucial to the successful diagnosis, treatment, and comprehensive management of various medical conditions (Yan et al. 2023). However, medical images are frequently transmitted over public networks, which are vulnerable to tampering, theft, and damage, resulting in severe economic losses and misdiagnoses. Security techniques for medical images can generally be classified into two categories: cryptography and information hiding. In cryptographic techniques, encryption mechanisms are frequently utilized to enhance the security of medical

Table 5 Quantum-enabled algorithms for medical image classification (data sources are expounded in Appendix)

Authors	Year	Tasks	Image modalities	Data sources	Methods	Metrics
Iliyasu and Fatichah (2017)	2017	Cervical cancer detection	Microscopic	Herlev dataset	QPSO	Precision = 0.85 F1-Score = 0.85 Recall = 0.86
Shi et al. (2019)	2019	Idiopathic pulmonary fibrosis prediction	HRCT	UCLA Computer Vision and Imaging Biomarkers	QPSO	Accuracy = 0.821 Specificity = 0.822 Sensitivity = 0.818
Acar and Yilmaz (2021)	2021	COVID-19 detection	CT	Hybrid dataset	QTL	Accuracy = 0.969 Precision = 0.974 Specificity = 0.976
Sengupta and Srivastava (2021)	2021	COVID-19 detection	CT	Hybrid dataset	QNN	Accuracy = 0.969
Al-Wesabi et al. (2023)	2022	Dystrophi-nopathies diag-nosis	MR	N/A	MOQTSO	Sensitivity = 0.984 Specificity = 0.958 Accuracy = 0.964
Sharma and Sharma (2022)	2022	Leukocytes classification	Microscopic	LISC dataset	QIBBA	Accuracy = 0.983
Umer et al. (2022)	2022	COVID-19 classification	X-ray	COVID-19 radio-graph	QTL	Accuracy = 0.990 Sensitivity = 0.987 Precision = 0.990
Amin et al. (2022)	2022	Brain tumor detection	MR	Kaggle/ BraTS 2020	QVC	Accuracy1 = 0.982 Accuracy2 = 0.997
Shahwar et al. (2022)	2022	Alzheimer's detection	MR	Alzheimer's dataset	CQNN	Accuracy = 0.97 Precision = 0.88 F1-Score = 0.86
Toledo-Cortés et al. (2022)	2022	Grading diabetic retinopathy	Color fundus photography	EyePACS	DQOR	Sensitivity = 0.915 Specificity = 0.846 AUC = 0.943
		Prostate cancer diagnostic	Microscopic	TCGA-PRAD	DQOR	Accuracy = 0.782
Houssein et al. (2022)	2022	COVID-19 prediction	X-ray	Private dataset	HQ-CNN	Accuracy = 0.822 Recall = 0.826 F1-Score = 0.888

Table 5 (continued)

Authors	Year	Tasks	Image modalities	Data sources	Methods	Metrics
Ovalle-Magallanes et al. (2022)	2022	Stenosis detection	X-ray	XCA	CQCNN	Accuracy = 0.918 Recall = 0.949 F1-Score = 0.918
Amin et al. (2022)	2022	COVID-19 classification	CT	UCSD-AI4H	QCNN	Precision = 0.96 Accuracy = 0.96 Recall = 0.95
Parisi et al. (2022)	2022	Parkinson's disease	Ultrasound	NewHandPD	QReLU	Accuracy = 0.83 Precision = 0.83 Recall = 0.83
		COVID-19 diagnosis	Ultrasound	POCUS 19	QReLU	Accuracy = 0.60 Precision = 0.87 Recall = 0.63
Landman et al. (2022)	2022	Medical image classification	X-ray	Retina-MNIST	QANN QONN	AUC = 0.77 Accuracy = 0.74
Azevedo et al. (2022)	2022	Breast cancer detection	Mammography	BCDR dataset	CQNN	Precision = 0.83 Recall = 0.80 Accuracy = 0.81
Ragab et al. (2022)	2022	COVID-19 diagnosis	X-ray	CXR image	QSGOA	Accuracy = 0.998 Sensitivity = 0.998 Precision = 0.998
Choudhuri and Halder (2023)	2023	Brain tumour classification	MR	Harvard	QCCNN	Precision = 1.00 Sensitivity = 0.976 Accuracy = 0.984
Tenali and Babu (2023)	2023	COVID-19 detection	X-ray	COVID-19 radiography	QCNN	Accuracy = 0.990 Precision = 0.988 Recall = 0.987
Qu et al. (2023)	2023	Breast cancer diagnosis	Ultrasound	Medminst	QCNN	Accuracy = 0.970 Precision = 0.974 Recall = 0.974
		COVID-19 diagnosis	CT	Covid-19 CT dataset	QCNN	Accuracy = 0.976 Precision = 0.953 Recall = 0.978
Zhao et al. (2023)	2023	COVID-19 prediction	CT	SARS-CoV-2	HQCNN	Accuracy = 0.993 Precision = 0.991
Darwish et al. (2023)	2023	Brain tumor detection	MR	BraTS 2019	QDA	Accuracy = 0.989 Precision = 0.951 Recall = 0.973

Table 5 (continued)

Authors	Year	Tasks	Image modalities	Data sources	Methods	Metrics
Mishra et al. (2023)	2023	Cervical cancer classification	Microscopic	N/A	QIWO	Accuracy = 0.990 Precision = 0.963 Recall = 0.967
Kulkarni et al. (2023)	2023	Pneumonia detection	X-ray	Pneumonia X-ray images	CQNN	Accuracy = 0.746 AUROC = 0.836
Hassan et al. (2023)	2023	Classification	CT/MR/X-ray	MedNIST	QCNN	Accuracy = 0.996 Precision = 0.997 Recall = 0.996
Deshmukh et al. (2023)	2023	Knee classification	MR	MRNet Dataset	Iqk-means	Accuracy = 0.937 Completeness = 0.932

images by converting the original medical image into meaningless information (Lin et al. 2022). Moreover, information hiding encompasses techniques such as watermarking (Shen et al. 2024) and steganography (Potdar et al. 2005). In contrast to the encryption of medical images, information hiding ensures security through the covert integration of paramount and specific data into the medical image. Traditional algorithms, such as chaotic mapping, may confer a measure of security; however, the pseudo-randomness they produce can be scrutinized and potentially compromised, exposing discernible security vulnerabilities (Li et al. 2012; Muhammad and Özkaynak 2019). In contrast, the pseudo-randomness generated by quantum-enabled algorithms (e.g., the quantum walk algorithm), which exploit the quantum properties of superposition and entanglement to increase the quantity and diversity of random samples, provides significantly greater security (Kadian et al. 2021; Joseph and Ali 2022). Additionally, the quantum no-cloning theorem and uncertainty principle make it exceedingly difficult for classical methods to crack encryption, thus enhancing anti-attack capabilities. Therefore, compared to traditional algorithms, quantum-enabled algorithms may offer higher security for applications in medical image processing. In this section, we discuss some recent quantum-enabled algorithms that have been applied to improve medical image security.

4.1 Medical image encryption

Medical image encryption techniques are utilized to protect the security of private information from patients by converting the original image into a meaningless, noise-like encrypted image that no longer contains any primitive, visually relevant information. Owing to quantum computing with superior performance and high security in solving specific problems compared to classical algorithms, many novel medical image encryption algorithms using quantum-enabled algorithms have been presented in recent years. As depicted in Fig. 9, we start with multiple quantum and quantum-inspired algorithms as sources and present the encryption/decryption (in the pink part below) and watermark embedding/extraction processes for medical images. In this subsection, we discuss and summarize the use of quantum-enabled algorithms for encrypting medical images.

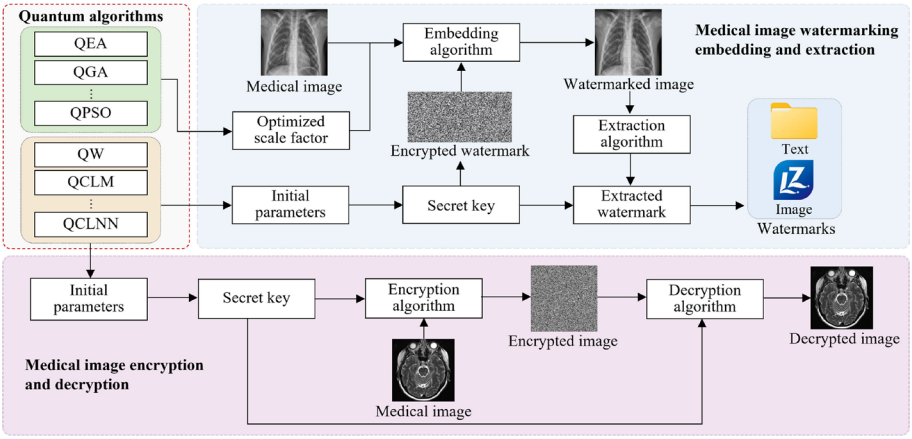


Fig. 9 General framework diagram of medical image encryption/decryption and watermark embedding/extraction using quantum-enabled algorithms

Pseudo-random numbers play an important role in the image encryption algorithm, which can effectively enhance the security of the encryption process. Herein, we provide an overview of related studies involving three quantum-enabled algorithms (i.e., QW, quantum cell neural network, and quantum chaotic map (QCM)) to generate pseudo-random numbers for medical image encryption. First, the QW algorithm has proven to be an excellent tool for designing modern cryptographic mechanisms owing to its ability to resist probable attacks and inherent nonlinear chaotic dynamic behavior. For example, EL-Latif et al. proposed an image encryption strategy based on controlled alternate quantum walk (CAQW) to preserve the privacy security of healthcare images in environments of the Internet of Things (IoT) (Abd EL-Latif et al. 2020). The proposed cryptosystem consisted of two phases (substitution and permutation), in which the CAQW on a circle was used to generate both keys, based on distinct initial key parameters. Similarly, to improve the security of the medical image, Abd-El-Atty et al. proposed a double medical image encryption scheme based on discrete QW and logistic map (Abd-El-Atty et al. 2023). In this cryptosystem, double plain medical images were divided into two distinct images, in which one with the highest four bits of each image was encrypted using the QW and the other with the lowest four bits was encrypted using the logistic map. A series of experiments were conducted on diverse medical images; the results demonstrate that the QW algorithm applied in medical image encryption is highly efficient in providing the security of medical images.

Second, a quantum cell neural network (QCLNN) can generate considerable pseudo-random sequences, which are further converted into security keys for image encryption (Di et al. 2017). For example, to protect medical image information, Zhou et al. proposed a selective image encryption method based on QCLNN and game theory (Zhou et al. 2020). In these encryption and decryption processes, the QCLNN was used to generate numerous secure and sensitive random sequences to scramble and diffuse the ROI. The experimental simulation results demonstrate that the QCLNN applied for medical image encryption can adequately protect medical image security against various attacks.

Finally, the quantum chaos system is derived from the classical logistic system and has high dimensions and complex dynamic behavior, which are utilized to enhance the randomness of cipher keys and their ability to defend against conventional attacks. Thus, Anitha et al. proposed an encryption scheme based on a quantum chaotic logistic map (QCLM) for the secure transmission of medical images (Anitha and Vijayalakshmi 2021). In this process, random network parameters were regarded as the initial conditions of the QCLM, which were used to generate highly complex cipher keys. The mammographic image analysis society (MIAS) dataset images served as the experimental test images to validate the effectiveness of the proposed algorithm. To provide better high-security performance for medical images, Yin et al. proposed a medical image encryption method that combines a modified quantum chaos system (MQCS) and a genetically simulated annealing PSO algorithm (Yin and Li 2021). In this scheme, the MQCS was efficiently employed to generate a significant number of key streams for medical image encryption, thereby bolstering its capability against histogram analysis, correlation analysis, differential attacks, and brute-force attacks. However, the key generated by the quantum chaotic system was independent of the plaintext, making it vulnerable to attacks. Alharbi et al. proposed a multistage encryption scheme using a linear feedback shift register (LFSR) and a quantum chaotic map (QCM) (Alharbi et al. 2022). In the encryption process, the keys were obtained using the LFSR and QCM to diffuse and substitute the original images, respectively. The experimental results indicated that the suggested encryption technique produced an encrypted image with high randomness and low correlation, which made the proposed encryption algorithm secure for data transmission.

A quantum-based key generator (QKG), on the other hand, was designed using Bell state measurement and has an advantage in the secure distribution of random key information against passive eavesdroppers. Lin et al. proposed intelligent symmetric cryptography for medical image encryption and decryption using the QKG and sine-power chaotic map. In this process, the QKG was used to generate two pairs of cipher codes to increase the complexity levels for changing the pixel values of medical images (Lin et al. 2021). To validate the effectiveness of the methodology, 100 chest radiograph images were selected from the NIH database; the results showed that the proposed intelligent symmetric cryptographic method exhibited greater confidence against the passive eavesdropper and the encrypted images showed a visual uncorrelation with the plain images.

4.2 Medical image watermarking and steganography

Digital watermarking techniques can be applied to medical images to achieve private data hiding (Elhadad et al. 2021), image authentication (Cedillo-Hernandez et al. 2020), and integrity verification (Huang et al. 2023). Recently, many researchers have actively explored the application of quantum-enabled algorithms in medical image watermarking to improve its original performance. In the upper blue part depicted in Fig. 9, we present the watermark embedding/extraction processes in quantum-assisted medical image processing. In this subsection, we recap relevant studies on medical image watermarking that utilize quantum-enabled algorithms from two perspectives.

One common application of quantum algorithms in medical image watermarking is the encryption of watermark information to enhance its security. For example, to protect electronic patient records (EPRs) and medical images, Dhall et al. introduced a multilayered secure authentic watermarking strategy designed for medical applications (Dhall and Gupta 2021). A quantum logistic map (QLM) was utilized to encrypt the compressed EPR before embedding the EPR into the medical image to generate a watermarked medical image. Furthermore, Sanivarapu et al. introduced a robust medical watermarking algorithm based on the discrete wavelet transform (DWT) and Schur decomposition for the authentication and tamper detection of medical images (Sanivarapu 2022). In this scheme, a quantum Hilbert image scrambling (QHIS) method was used to encrypt multiple watermark images, ensuring high security during the transmission of watermarked images. Recently, Yan et al. proposed a multi-watermarking strategy that utilized QW and Brain Storm Optimization (BSO) algorithms to verify the integrity and authenticity of medical images in the Internet of Medical Things (IoMT) environment (Yan et al. 2022). To increase both the security and sensitivity of the watermark, the QW algorithm was employed to produce many pseudo-random numbers for the encryption of the logo image. Then, the encrypted and compressed watermarks were embedded in the ROI and RONI of medical images, respectively. More importantly, several experiments were conducted using various cover medical images and watermark information, consistently demonstrating the ability of quantum algorithms to provide robust security and authenticity in healthcare applications.

Robust medical image watermarking techniques generated watermarked images along with contradictions in terms of robustness and imperceptibility. Another typical application of quantum algorithms in medical image watermarking is used to trade-off between the robustness and imperceptibility of watermarked images. In the early stages, Soliman et al. proposed an adaptive medical image watermarking scheme based on the QPSO algorithm to safeguard the copyright and authentication of medical images (Soliman et al. 2012). The QPSO algorithm was performed to make the trade-off between the imperceptibility

and robustness of watermarked medical images. Subsequently, Samanta et al. proposed a hybrid medical watermarking approach for the authentication of biomedical content using discrete cosine transformation (DCT), DWT, and singular value decomposition (SVD) (Samanta et al. 2017). In this process, the GA, QGA, and quantum-inspired evolutionary algorithm (QEA) were employed to individually select an optimal scaling factor to embed the watermark information. Several experimental results showed that the proposed method established an accurate authentication system for information exchange with high robustness, imperceptibility, and payload. Moreover, the results proved the advantages of both QEA and QGA over the GA in terms of imperceptibility and robustness. In summary, to improve the superiority of medical image watermarking techniques, quantum algorithms applied to medical image watermarking can not only be used to encrypt watermarks to enhance the security of watermark information but can also achieve the balance between the imperceptibility and robustness of watermarked images.

Steganography is another information-hiding technique that can embed data into media signals to achieve covert communication. Note that the medical image steganography technique conceals secret messages in a way that aims to avoid detection, interception, or decoding (Wang et al. 2023). In addition to the difference between digital watermarking in terms of purpose and applications, steganography requires a higher payload capacity to effectively camouflage secret messages. Similarly, quantum algorithms can be used to encrypt the embedded data to enhance the security of the stego image during the transmission process. For example, Abd-El-Atty proposed a novel image steganography strategy to provide high security to medical data (Abd-El-Atty 2023). The QW algorithm and 3D chaotic system were utilized to operate the PSO algorithm, in which the generated velocity sequence was used to substitute the confidential data, and the position sequence was used to select the position in the hosting image employed to host the substituted confidential data. The proposed method indicated that the application of quantum algorithms to medical image steganography has tremendous potential to provide high levels of security for the stego image.

Some typical information-hiding techniques using quantum algorithms, including the aforementioned studies, are presented in Table 6. The evaluation metrics of each scheme were provided to validate the effectiveness of quantum algorithms in medical image security tasks. Unlike encryption techniques, the capacity to embed information in watermarking and steganography plays a crucial role in their implementation and evaluation; hence, the embedding capacity in these studies is outlined. Similar to the discussions on disease diagnosis, the table displays the results of various metrics for each scheme, primarily for intuitive reference rather than direct technical comparison, considering the potential variations in experimental settings among the cited studies.

5 Discussions and conclusions

Quantum computing has emerged as a robust and influential tool for the efficient and secure processing of information. Notably, quantum algorithms have tremendous potential to outperform classical algorithms in solving specific problems, and quantum mechanical effects can significantly enhance the computing speed and provide tamper-proof security. In recent years, numerous researchers have dedicated their studies to applying quantum-enabled algorithms in the field of medical image processing. In this review, we provide a comprehensive overview of the advancements in quantum-assisted medical image processing for

Table 6 Quantum algorithms for medical image encryption, watermarking, and stegonagraphy

Applications	Authors	Year	Image types	Methods	Image size/embedding capacity	Metrics
Watermarking	Soliman et al. (2012)	2012	Grayscale	QPSO	512 × 512/32 × 32	PSNR = 54.64
	Samanta et al. (2017)	2017	Grayscale	QGA/QEA	N/A	PSNR = 46.10 NCC1 = 0.965 NCC2 = 0.939 NCC3 = 0.947
	Dhall and Gupta (2021)	2021	Color/grayscale	QLM	256 × 256/150 bytes	PSNR = 51.68 MSE = 0.333 NCC = 0.998
	Sanivarapu (2022)	2022	Grayscale	QHIS	256 × 256/64 × 64	PSNR = 34.90 NCC = 0.998
	Yan et al. (2022)	2022	Grayscale	QW	512 × 512/359 characters	PSNR = 52.16 NCC = 1.00
Steganography	Peng et al. (2019)	2019	Color/grayscale	QW	2-bit/pixel	PSNR = 48.60 UIQ[a] = 0.972 SSIM = 0.996
	Abd-El-Atty et al. (2020)	2020	Color/grayscale	CAQW	256 × 256/2-bit/8-bit	PSNR = 44.23 UIQ = 0.948 SSIM = 0.949
	Abd-El-Atty (2023)	2023	Color/grayscale	QW	512 × 512/2-bit/8-bit	PSNR = 44.10 UIQ = 0.967 SSIM = 0.988
Encryption	Abd EL-Latif et al. (2020)	2020	Color/grayscale	CAQW	256 × 256	NPCR[b] = 0.996 IE = 7.998
	Zhou et al. (2020)	2020	Grayscale	QCLNN	512 × 512	UACI[c] = 0.333 NPCR = 0.999 IE[d] = 7.999
	Lin et al. (2021)	2021	Grayscale	QKG	1024 × 1024	UACI = 0.319 NPCR = 0.994 IE = 7.553
	Anitha and Vijayalakshmi (2021)	2021	Grayscale	QCLM	256 × 256	NPCR = 0.992
	Yin and Li (2021)	2021	Grayscale	MQCS	512 × 512	UACI = 0.333 NPCR = 0.996 IE = 7.997
	Alharbi et al. (2022)	2022	Color	QCM	512 × 512	UACI = 0.334 NPCR = 0.992 IE = 7.999
	Abd-El-Atty et al. (2023)	2023	Grayscale	QW	512 × 512	UACI = 0.334 NPCR = 0.996 IE = 7.429
	Rehman et al. (2023)	2023	Color	QW	256 × 256	IE = 7.999

[a]UIQ: Universal Image Quality; [b]NPCR: Number of Pixels Change Rate; [c]UACI: Unified Average Changing Intensity; [d]IE: Information Entropy

disease diagnosis and medical image security. Our efforts in the former part of this review aim to provide readers with a concise compendium of the current state of the field, while in the rest of this review, we further analyze open challenges and future directions existing in this field.

5.1 Analytical discussions

Many related studies on quantum-enabled algorithms applied to disease diagnoses and security technologies in medical image processing have recently been proposed, with the purpose of improving overall performance. However, there remain problems and challenges in these studies that need to be solved urgently. Herein, we provide thorough discussions and future perspectives for quantum-assisted medical image processing. We hope that discussions on this development will arouse the interest of more researchers and accelerate efforts toward achieving more sophisticated applications in medical image processing using quantum-enabled algorithms.

5.1.1 Limitations in quantum-assisted medical image processing

Although many researchers have sought to utilize quantum-enabled algorithms to address various challenges in medical image processing, these algorithms and their applications still have limitations that require further attention and resolution. Specifically, the following limitations in quantum-assisted medical image processing warrant discussion:

- Numerous studies have confirmed that quantum-enabled algorithms can achieve better experimental results than traditional algorithms in the disease diagnosis process. Quantum machine learning algorithms, in particular, show significant advantages in addressing specific issues within applications such as medical image segmentation and classification. However, due to the current immaturity of the hardware supporting these algorithms, quantum machine learning algorithms remain limited when simulating and processing high-dimensional medical data on classical computers. Therefore, traditional machine learning algorithms may be more effective than quantum machine learning when dealing with complex medical tasks. Additionally, in the field of medical image security, QW algorithms can serve as pseudo-random number generators, producing highly random and unpredictable sequences for medical image encryption. However, QW involves complex quantum operations and high computational complexity. Since most QW algorithms are currently simulated on classical computers, their resource consumption increases significantly when processing large-scale medical data, and their performance is limited by the capabilities of existing computing systems. Besides its many applications in medical image security, QW has also been applied by researchers to medical image segmentation tasks, but the experimental results were less than satisfactory (Krok et al. 2019). In summary, while quantum-enabled algorithms have advantages in addressing specific medical image processing tasks, they still have certain limitations in more general tasks.
- Medical image processing serves as a fundamental task with many practical applications in areas such as computer-aided diagnosis and surgical navigation systems. However, since quantum-assisted medical image processing is still in its early stages, most existing studies have been conducted through simulation experiments using publicly

available real-world data sets on classical computers to validate the effectiveness of the proposed solutions. Additionally, the current hardware and quantum algorithm implementations are not yet mature enough to be directly deployed in actual medical applications. As a result, the real scenarios for applying quantum-assisted medical image processing techniques remains limited at present. Nonetheless, the under-development Cleveland Clinic-IBM Discovery Accelerator is an example of a project that attempts to overcome the quantum healthcare services on commodity hardware challenges (Ur Rasool et al. 2023). Medical image processing demands efficient and real-time capabilities in clinical applications, but existing research in quantum-assisted medical image processing struggles to meet these requirements in practical clinical environments.

- As discussed earlier, quantum-enabled algorithms encompass both quantum and quantum-inspired algorithms. However, in the field of medical image processing, the application of quantum-inspired algorithms currently far exceeds that of quantum algorithms. This is likely because designing effective quantum algorithms is inherently a challenging and arduous task at the current stage. Additionally, existing well-known quantum algorithms, such as Grover's algorithm, are difficult to be simulated and implemented in medical image processing applications. In the absence of viable quantum algorithms and the requisite supporting hardware, the purported quantum performance and advantages in medical image processing tasks are actually limited. Furthermore, even the existing quantum-enabled algorithms in medical image processing are relatively narrow in their applications, being mostly confined to specific tasks rather than broadly applied across the field. For example, a generative adversarial network is used as the main learning network to perform the encryption and decryption of medical images as a style transfer task (Ding et al. 2020). However, its quantum counterpart has not yet been applied to medical image encryption. This may require the involvement of more researchers interested in this emerging field.

5.1.2 Open questions in quantum-assisted medical image processing

A few open questions still need to be solved within the context of the advances made in quantum-assisted medical image processing. In particular, these concerns are discussed together with the corresponding possible solutions as follows:

- Although the definitions of quantum algorithms and quantum-inspired algorithms are clear, in simple terms, quantum algorithms run on a realistic model of quantum computation, while quantum-inspired algorithms integrate quantum elements into their traditional counterparts and run on classical computers. However, since specialized quantum computing devices have not yet emerged, quantum algorithms are still a simulated process in medical image processing applications. As a result, in current medical image processing research, many studies have not clearly differentiated the implementation process and the roles played by quantum and quantum-inspired algorithms. This ambiguity is very inappropriate because the efficiency and advantages introduced by these two types of algorithms differ. For example, QW exploits the unique properties of quantum systems to potentially achieve computational advantages, which can serve as an efficient generator of pseudo-random numbers to enhance the security of medical images (Abd-El-Atty et al. 2023). Although QGA incorporates principles or ideas inspired by quantum computing to improve the performance of the traditional genetic algorithm, with the aim of exploring the

search space more efficiently and potentially finding superior solutions to optimization problems in the medical field (Hilali-Jaghdam et al. 2020). Simultaneously, the feasibility of implementing medical image processing on future dedicated quantum computing devices also differs, that is, quantum algorithms may perform and be implemented more effectively on future quantum computers, thereby offering innovative solutions to real-world problems. This review has made significant efforts in this regard to mitigate conceptual confusion and alleviate technical misconceptions.

- The exact level of performance improvement caused by the application of quantum-enabled algorithms in the field of medical image processing remains sufficiently unclear. First, as previously mentioned, quantum and quantum-inspired algorithms can be applied to various tasks in medical image processing, such as edge detection and image segmentation. Although these quantum algorithms exhibit diversity in their implementation, it is important to note that their functionality tends to be relatively limited and may unnecessarily play significant or pivotal roles in the current research landscape. For example, some quantum-enabled algorithms are solely utilized for the parameter optimization of neural network models or original image enhancement in related studies of medical image segmentation tasks. It is unclear how quantum technology has improved the performance of the final segmentation results in terms of accuracy evaluation indicators. However, quantum algorithms have been previously applied in (nonmedical) image segmentation as key solutions. For example, Krok et al. proposed an image segmentation method using continuous time QW (Krok et al. 2019). Therefore, exploring how to further apply such algorithms in medical image analysis and achieve crucial breakthroughs in core technologies is worth studying.
- Although we explore and discuss the applications of quantum-enabled algorithms in medical image processing in detail, most of these algorithms aim to improve the original counterpart classical algorithms to offer the potential for faster computation, increased accuracy, or the ability to handle complex data structures efficiently. However, the fundamental principles of quantum mechanics, such as superposition and entanglement, have not been fully utilized in the field. This could be attributed to two underlying factors, potentially contributing to this phenomenon. First, researchers in this field are mainly focused on computer scientists, with a greater focus on the design and evolution of engineering algorithms, as well as the continuous improvement of performance and metrics. Second, the design of algorithms, model construction, and specification of protocols that incorporate quantum attributes involve intricate quantum mechanical processes. This requires a substantial background in terms of physics and practical experience. Therefore, unfamiliarity with these unknown advantages leads to an inherent lack of intrinsic motivation for scientists currently working in this field. Thus, it is imperative to involve physicists who are interested in medical image processing to truly achieve disruptive performance advantages and harness the full potential of quantum properties in addressing unsolvable problems within classical domains. For example, quantum algorithms have been directly applied in the medical domain for surgical navigation. Undoubtedly, an alternative that could facilitate revolutionary advancements in the field is the design of more sophisticated quantum algorithms or the incorporation of additional quantum properties, with the help of physicists.

5.1.3 Future directions for quantum-assisted medical image processing

To stimulate research interest and drive advancements in the field, we explore short-, mid-, and long-term future directions in the application of quantum and quantum-inspired algorithms to medical image processing, as depicted in Fig. 10. These future directions and their corresponding points of attention are elucidated and summarized as follows:

- Short-term goal: The development and applications of quantum-enabled algorithms encompass numerous domains in medical image processing, as presented in Fig. 1a, yet the progress in these directions is uneven. In particular, within the domain of medical image processing tasks, several areas have not yet been thoroughly explored and therefore require further investigation. For instance, research on the use of quantum and quantum-inspired algorithms to address related issues in the field of medical image fusion, registration, and reconstruction is currently lacking. Hence, in addition to refining and improving existing algorithms and applications, there exists a pressing need to place greater emphasis on the development of cutting-edge algorithms that remain unexplored to solve new medical image tasks in this field. In this matter, we must pay attention to the following points. (1) Practitioners always strive for continuous improvement and progress in experimental performance. It is noteworthy that the development of a novel quantum algorithm for problem solving, even if its performance is not optimal, still deserves recognition and appreciation. (2) It is imperative to explicitly clarify the roles played by the adopted quantum and quantum-inspired algorithms in the implementation process of medical image processing so that the specific contributions of these algorithms to this task can be fairly evaluated. (3) An objective comparison of similar quantum algorithms is essential to facilitate the advancement and development of this field. For example, dedicated datasets and specifically tailored estimating metrics should be established for specialized quantum algorithms.
- Medium-term goal: With the maturation of algorithmic research in this domain and the growing involvement of researchers, it is necessary to progressively establish specialized quantum computing modules that can be effectively harnessed for medical image processing to expedite the systematic and standardized advancement of the field. As an illustration, the utilization of quantum adders in cryptography enables the generation and manipulation of quantum states, thereby facilitating secure communication and data transmission. Simultaneously, it is crucial to develop and integrate toolkits or packages (such

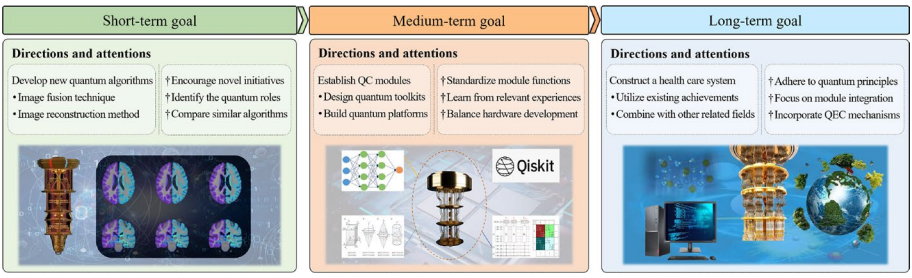


Fig. 10 Short-, mid-, and long-term research directions and consideration notes for quantum-assisted medical image processing (QC and QEC indicate quantum computing and quantum error correction, respectively)

as QLib in MATLAB (Machnes 2007)) specifically tailored for use on existing medical image processing platforms. Furthermore, the development of dedicated simulation platforms, such as International Business Machine (IBM) Qiskit, can be undertaken to validate quantum algorithms in the domain of medical image processing. The above strategies can effectively prevent duplicate design of algorithms that achieve the same function, improve the correctness and efficiency of algorithm design and experimental verification, and thus enable researchers to focus more on designing more advanced algorithms. In this respect, several key aspects require attention: 1) Standardizing the encapsulated functionalities is crucial when designing computational modules. In essence, it is crucial to strike a balance in the design of computational modules. This involves avoiding excessive functionality that compromises universality and applicability, as well as insufficient functionality that hampers module utilization efficiency. Additionally, to facilitate customized utilization based on specific problem requirements, the designed modules should incorporate specific parameters and interfaces. 2) We should fully consider the integration of research directions related to quantum-assisted medical image processing, such as the rapidly growing field of quantum image processing (Yan et al. 2021). When implementing the aforementioned modules or toolkits, we can draw on the experiences gained from the relevant fields and share the achievements in this field to promote mutual rapid development. 3) The inherent interdependence between hardware and software represents a fundamental aspect that requires careful consideration within computing systems. In other words, the hardware infrastructure of quantum computing serves as the fundamental underpinning upon which software implementations depend and operate. In the process of designing the aforementioned software features, it is crucial to simultaneously consider their integration and harmonization with upcoming specialized quantum hardware architectures.

- Long-term goal: Quantum computing exhibits broad applications in medical image processing and in the field of healthcare, as analyzed in Table 1. The long-term development strategy in this field should aim to establish a comprehensive quantum-enabled ecosystem for healthcare, encompassing a wide array of medical services and treatments. If possible, we may fully utilize the advantages of quantum mechanics, including quantum teleportation, to achieve accurate detection and secure transmission of medical images. As discussed in the Introduction section, quantum states of medical images may be generated and stored using quantum-correlated imaging techniques, and these states can be further manipulated using quantum gates and control operations, thereby ultimately achieving the objectives of medical image processing in the form of quantum transformations. From this perspective, advances in the other two classifications mentioned in the Introduction section, that is, quantum medical image processing and quantum medical imaging, are also very urgent currently. Consequently, it is imperative to consider the following aspects. 1) Throughout the entire system construction, it is paramount to fully harness existing protocols and technologies, such as quantum key distribution, and any endeavor should not disobey the fundamental principles of quantum mechanics. 2) It is crucial to prioritize the interconnection of multiple modules, with special emphasis on the design of interfaces for quantum-to-classical and classical-to-quantum information conversion. 3) Given the influence of factors, such as quantum noise, it is imperative to fully integrate quantum error correction mechanisms into the construction of the entire system to ensure the precise implementation of medical image processing.

5.2 Concluding remarks

Medical images are commonly used to diagnose various diseases, with the ongoing progress of medical devices and technologies. However, the high volume of images that radiologists must diagnose results in an inherent risk of missed diagnosis. In addition, the secure and efficient storage, transmission, and analysis of medical images are vital for patients, doctors, and other medical personnel. Quantum algorithms exploiting the principles of quantum mechanics are widely used to design algorithms that outperform, enhance, or complement classical algorithms. Recently, quantum-enabled algorithms utilized in the field of medical image processing have demonstrated their effectiveness in assisting physicians and relevant personnel with highly accurate disease diagnosis and reliable privacy protection.

In this review, we first discussed the progress in disease diagnosis and security techniques in quantum-assisted medical image processing in recent years. The fundamentals of quantum computing and the technical details of some known quantum and quantum-inspired algorithms were described organically. Related studies on preliminary processing, segmentation, and classification of medical images in disease diagnosis using quantum-enabled algorithms were systematically outlined and analyzed. Furthermore, we comprehensively reviewed related studies on medical image-based cryptography and information-hiding techniques using quantum-enabled algorithms. Finally, open problems and future perspectives were discussed to further explore the quantum advantages of these algorithms and promote the development of this field.

Marie Curie stated that “We must believe that we are gifted with something, and that this thing, at whatever cost, must be attained." Despite the slow and challenging progress in developing large-scale and general-purpose quantum computers, this will not hinder the exploration and design of quantum and quantum-inspired algorithms in various applications, including medical image processing. Hopefully, the efforts made in this review will stimulate further interest in the pursuit of more advanced quantum-enabled algorithms to solve key choking problems and experimental validations of available technologies in medical image processing. In addition, we are convinced that continuous research and collaboration between quantum scientists and medical experts are crucial to unlock the full potential and benefits of this emerging subarea.

Appendix 1

Tables 7 and 8 are provided to aid readers in understanding discussions in the main body of this review.

Table 7 Abbreviations of quantum and quantum-inspired algorithms used in this review

Acronym	Description	Acronym	Description
PSO	Particle swarm optimization	QPSO	Quantum-inspired particle swarm optimization
GA	Genetic algorithm	QGA	Quantum genetic algorithm
QINN	Quantum-inspired neural network	QIBF	Quantum-inspired bilateral filtering
QITF	Quantum-inspired thresholding function	QMFT	Quantum matched-filter technique
QWTF	Quantum wavelet transform filtering	IQC	Improved quantum-Canny
QuLBP	Quantum local binary pattern	QSP	Quantum signal processing
QE-LBP	Quantum entropy-local binary pattern	GQAOA	Gaussian quantum arithmetic optimization
SCQPSO	Partitioned and cooperative quantum particle swarm optimization	QMPA	Quantum-inspired marine predators algorithm
QIFA	Quantum-inspired firefly algorithm	QIMGA	Quantum-inspired modified genetic algorithm
FFQO	Fast forward quantum optimization	QIWO	Quantum invasive weed optimization
QIS-Net	Quantum-inspired self-supervised neural network	LQCNN	Lightweight quantum and convolutional neural networks
QIBBA	Quantum-inspired binary bat algorithm	MOQTSO	Multi-objective quantum tunicate swarm optimization
QVC	Quantum variational classifier	DQOR	Deep quantum ordinal regressor
QONN	Quantum orthogonal neural networks	Iqk-means	Quantum k-means
CQNN	Classical-quantum neural network	CQCNN	Classical-quantum convolutional neural network
QCCNN	Quantum-classical convolutional neural network	QTL	Quantum transfer learning
QNN	Quantum neural network	QReLU	Quantum rectified linear unit
QW	Quantum walk	CAQW	Controlled alternate quantum walk
QCLNN	Quantum cell neural network	QCLM	Quantum chaotic logistic map
MQCS	Modified quantum chaos system	QCM	Quantum chaotic map
QKG	Quantum-based key generator	QHIS	Quantum Hilbert image scrambling
QEA	Quantum-inspired evolutionary algorithm	QLM	Quantum logistic map

Table 8 Datasets used in quantum-assisted medical image processing and their available URLs

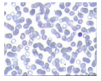
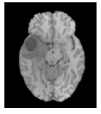
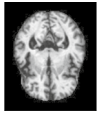
Image type	Data source	Total number	Sample	Available URL
Cervical smear images	Herlev dataset	917		https://mde-lab.aegean.gr/index.php/downloads/
Blood smear images	LISC dataset	400		http://users.cecs.anu.edu.au/hrezatofighi/Data/Leukocyte%20Data.htm
Chest images	COVID-19 radiography database	33920		https://kaggle.com/tawsifurrahman/covid19-radiography-database
Brain images	BraTS 2020 dataset	51925		https://www.kaggle.com/datasets/awsaf49/brats2020-training-data
Brain images	Alzheimer's dataset	6400		https://www.kaggle.com/tourist55/alzheimers-dataset-4-class-of-images
Prostate tissue images	TCGA dataset	400		http://cancergenome.nih.gov/
Eye fundus images	EyePACS dataset	65617		https://www.kaggle.com/c/diabetic-retinopathy-detection/data
COVID images	UCSD-AI4H dataset	9548		https://github.com/UCSD-AI4H/COVID-CT/tree/master/Images-processed

Table 8 (continued)

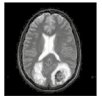
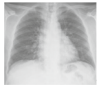
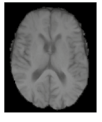
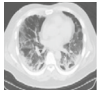
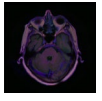
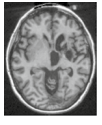
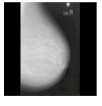
Image type	Data source	Total number	Sample	Available URL
COVID images	POCUS 19 dataset	53		https://github.com/jannisborn/covid19_ultrasound
Brain images	Harvard dataset	130		https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1738344/
Chest images	CXR image dataset	3224		https://github.com/ieee8023/covid-chestxray-dataset
Chest images	Pneumonia X-ray images dataset	4192		https://www.kaggle.com/pcbreviglieri/pneumonia-xray-images
Brain images	BraTS 2019 dataset	44175		https://www.kaggle.com/datasets/aryashah2k/brain-tumor-segmentation-brats-2019/code?resource=download
Medical images	MedNIST dataset	58954		https://www.kaggle.com/datasets/andrewmvd/medical-mnist
COVID images	SARS-CoV-2 dataset	2482		https://www.kaggle.com/plameneduardo/sarscov2-ctscan-dataset

Table 8 (continued)

Image type	Data source	Total number	Sample	Available URL
Chest images	COVID-19 radiography database	5856		https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database
COVID images	COVID CT dataset	349		https://github.com/UCSD-AI4H/COVID-CT/tree/master/Images-processed
Brain images	LGG Segmentation dataset	7858		https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation
Brain images	ATLAS R2.0 dataset	655		https://atlas.grand-challenge.org/
Breast images	MIAS dataset	322		http://peipa.essex.ac.uk/info/mias.html

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Consent to participate All authors read and agreed to participate in the final manuscript.

Consent for publication All authors agreed to publish this paper, if accepted.

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